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Introducing the Pattern Book

Welcome Jugglers and Friends

Welcome to the *Madison Area Jugglers' Pattern Book*, a collection of club-passing patterns for intermediate to advanced passers. These patterns are part of our juggling repertoire. Many we learned or adapted from other jugglers; we include them because it's fun trying to juggle them. But very many of the patterns in this book were born in Madison, Wisconsin, conjured up at Thursday/Sunday juggling sessions or at 47th Annual Madfests, annual events since 1992. The book offers some record of the evolution of club passing in Madison.

All jugglers are welcome to play with, use, distribute, extend, and otherwise enjoy all the club-passing patterns described in *Madison Area Jugglers' Pattern Book, Version 4,* which is licensed by its creators, the Madison Area Jugglers, under the Creative Commons Attribution 3.0 Unported License . You may access the book online, or check the latest updates at our GitHub repository.

If you have questions, comments or suggestions relating to this book, please contact the Madison Area Jugglers. Where possible we have given credit to the creators of the patterns described here, though for many these origins were unknown to us. If we have erred in our attributions, please let us know.

This collection is a living document. New contributions are welcome for the next version, so please join the team and provide some pattern descriptions. ***give an easy instruction on how to add***

About the Project

Originally we wrote down the patterns just to save time arguing over exactly how we did the pattern last time. (Don't worry...there's still plenty of arguing, even with the book.) We were pleasantly surprised to find other people found our book helpful too! The pattern book can be used as a club passing guide or just a library of the patterns that

we originated or find interesting. If you're from Madison, then it's a handbook to what we do twice a week, on Thursdays and Sundays, and if you don't know the patterns already then you probably will. This book is not intended to be an encyclopedia of all possible passing patterns (that would be a big book)! In fact, we don't include patterns just because they are *theoretically* possible; we include them if we have done them!

The Madison Area Jugglers (MAJ) have been attempting, inventing, and recording juggling patterns for longer than most of us can remember. In 1997, when our memories were already getting strained by the variations, Michael Newton (aka Prof Mike) compiled our first *Pattern Catalogue* to record the patterns for four or more jugglers. This morphed into the first *Pattern Book* by him in January 1999, but the hand-written collection (*Version 0.1*) was in a pretty limiting format. By October of that year, Steve Otteson created the first computerized collection by reformatting and expanding Mike's hand-written notes. Melonhead still carries around a tattered printout of *Version 1.0*. Steve made a few version updates, for example adding a bunch of extra-club patterns by Peter Kaseman and Luke Emery; the unusual computer format of Steve's version and his departure from Madison slowed down subsequent updates, even though the MAJ continued to create new patterns at a steady rate. The next big contribution was Luke Emery's version (*Version 2.0*, 2012). Though it was not widely distributed, Luke's version contained all kinds of new additions: more patterns for all numbers of jugglers, more on site-swap, more on specialized extra-club patterns, and tips for creating and doing patterns.

In 2015 Matt Hemke and Evan Driscol created a Github repository for the project, converted Luke's version into a repo, and updated the collection further. Chan Wilson, Jeff Adamus, and Melonhead began tracking the patterns we do at Thursday and Sunday MAJ practices. It's an amazing collection, and for those who remember the code words, like *1-5-1 Foie Graham* or *4-5 Twister*, it's enough to figure out what we really did. But most of us forget the code words after a while, except for the frequently juggled ones, and this was an impetus for the *Version 3.0* update by Prof Mike in 2018. Since then, we've suffered through two years of COVID restrictions and two lost Madfests (only 1 lost if you count spring ‘22). But we kept juggling and discovered some cool new patterns that work well with face masks! Nathan Nard's rejuvenating and inventive contributions helped to launch *Version 3.1* in 2023.

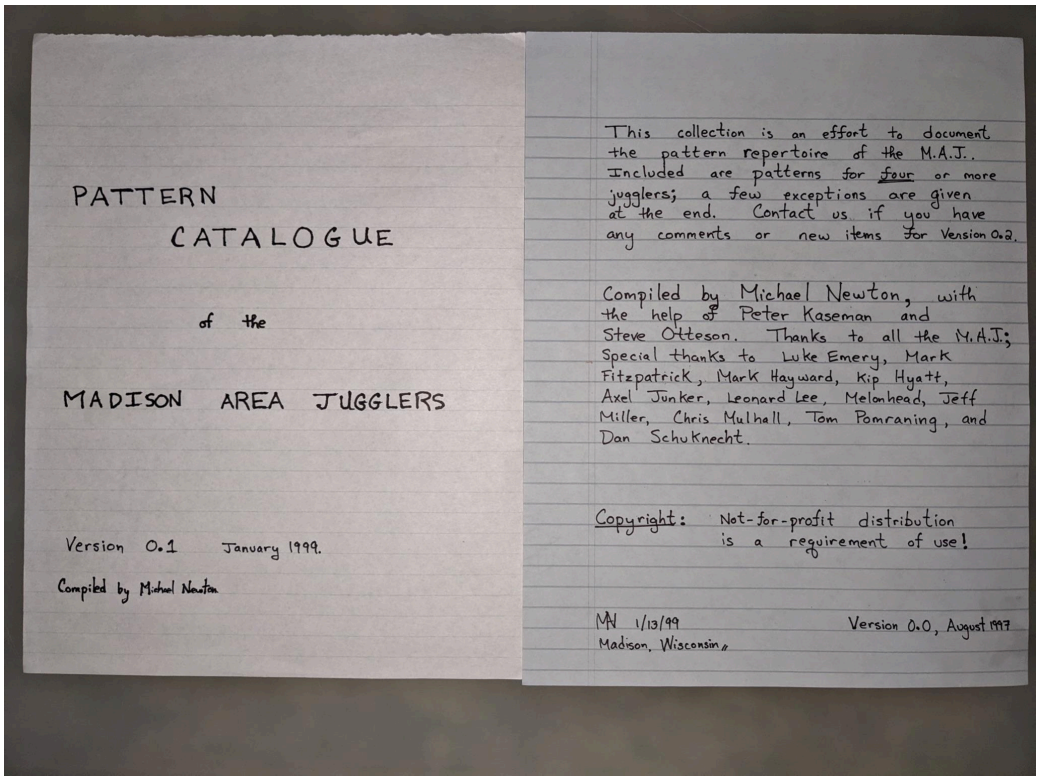
All along the editors have had substantial input from the Madison Area Jugglers and friends, and we are indebted to them for helping enhance this collection. Club passing is a social pastime that needs a group of similarly inclined jugglers. Many of these patterns are inspired by other jugglers and patterns or created in collaboration with

other jugglers. We hope that these patterns will help inspire new club passing ideas and help more jugglers all over the world enjoy club passing!

The Madison Area Jugglers

An intro paragraph or two on the MAJ

Version History



PATTERN
CATALOGUE

of the

MADISON AREA JUGGLERS

Version 0.1 January 1999.

Compiled by Michael Newton

This collection is an effort to document the pattern repertoire of the M.A.J. Included are patterns for four or more jugglers; a few exceptions are given at the end. Contact us if you have any comments or new items for Version 0.2.

Compiled by Michael Newton, with the help of Peter Kaseman and Steve Otteson. Thanks to all the M.A.J.; Special thanks to Luke Emery, Mark Fitzpatrick, Mark Hayward, Kip Hyatt, Axel Junker, Leonard Lee, Melonhead, Jeff Miller, Chris Mulhall, Tom Pomraning, and Dan Schuknecht.

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MN 1/13/99 Version 0.0, August 1997
Madison, Wisconsin

* Madison, Wisconsin, August 1997 Version 0.0
* Madison, Wisconsin, January 1999 Version 0.1
* Madison, Wisconsin, October 1999 Version 1.0
* Madison, Wisconsin, January 2001 Version 1.1
* Madison, Wisconsin, March 2001 Version 1.2
* Madison, Wisconsin, January 2012 Version 2.0

* Madison, Wisconsin, January 2018 Version 3.0
* Madison, Wisconsin, January 2023 Version 3.1

Updates in *Version 3.1*:

We added a bunch of new patterns (e.g., *laxatives, the pollinator, the zigzag, sneaky-snake, that's-a-really-good-one, fully-loaded-blast, Mel's-dream*, and more). We added more on big patterns, including on the super-fun 24-person *hex-of-clovers*. At recent practices we've experimented a lot with colors, in which jugglers each start with a distinct set of clubs, and we see how long it takes to move the colors around. This version has some details on the substructure of a few of our cool patterns.

Thanks to Jeff Adamus for posting YouTube recordings of a bunch of our patterns; we've added links ***where* (Thanks to Jeremy, Jessie, and ? others for great camera work!) . *other animations?***

Like all earlier versions, the current version contains mistakes and unfortunate omissions, which we have yet to discover and for which the Editors take responsibility. We hope that future versions will correct these shortcomings.

Updates in *Version 4.0*

****google doc version..prep for real book…new patterns …more uniform pics/tabular descriptions; pattern metrics** [lead by MAN]; still missing manipulator patterns…waiting for the right expertise to join the editors!**

The Editors



Michael Newton (aka Prof Mike), Luke Emery, and the Madison Area Jugglers, especially (in alphabetical order): Jeff Adamus, Nick Aikens, Josh Casey, Steve Clipson, Evan Driscol, Mark Hayward, Matt Hemke, Peter Kaseman, Melonhead, Nathan Nard, Graham Paasch, Steve Otteson, and Chan Wilson. You can be an editor too by working on the next version!

*****SEE PART 1 for PRELIMINARIES AND ALL THE STUFF THROUGH 5-PERSON PATTERNS*****

Six Person Patterns

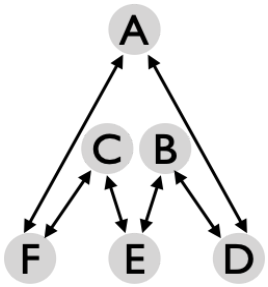
Stationary Six Person Patterns

1-2-3 Apollo

Nice and easy for six jugglers

Beat		
	1	2
A	D	F
B	E	D
C	F	E
D	A	B
E	B	C
F	C	A

4-count base



Everyone is on 4-count passing to 2 other jugglers over the pattern cycle, with exchanges going between pairs (i.e., pass to and receive from the same person.) Positions differ a bit, owing to the passing distances and the turning required. **A** is most difficult, followed by **F** and **D**, followed by easiest spots **B**, **C**, and **E**, which are ideal for first-time passers.

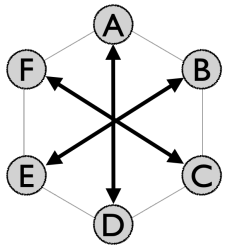
It keeps people engaged to rotate jugglers to new positions after a successful run; that way everyone gets a feel for each of the spots (and stays more alert!).

Pattern by the MAJ, we think.

Metrics: J=6, CJ=3, pH=1, cycle=2 on a 4-count base, OJ=2, PBJ=¼, and motion=0. To a first approximation, R=1 (i.e., there's just one role), but in fact **A** ought to oscillate left-right a bit to simplify passes to **F** and **D,** and juggler spin and passing distances vary. To square up to their partner on each beat, **A**, **B**, **C**, and **E** each swivel through an angle of measure about 4ϕ per cycle, where $sin(\phi) = 1/\sqrt{5}$, assuming the distance from **A** to **E** is about the same as the distance from **F** to **D**. That gives a pattern average spin $8\phi/3 \simeq 1.24$ rad = 71°.

Hex

The six person box.



	Juggler					
Beat	**A**	**B**	**C**	**D**	**E**	**F**
1	**D**			**A**		
2		**E**			**B**	
3			**F**			**C**

Jugglers are on 3-count (standard) or 6 count (slow), each to one partner.

Hex is a pleasant pattern for six, in which everyone is doing the same thing (R=1), and there's neither motion nor any spin. It's easier than the *Feast* or the *Apollo*, but it does require good timing to avoid collisions.

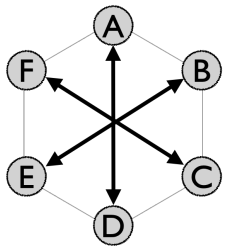
Metrics: J=6, CJ=3, pH=1, OJ=1, R=1, cycle length=3 on 1-count base (standard) or 2-count base (slow), PBJ=⅓ (standard), ⅙ (slow). Motion and spin are both 0.

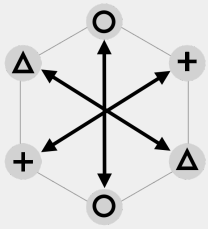
Once performed with Laptop Dave in the center wearing a helmet! In principle versions can be done with simultaneous throws on various counts, as with 4-person *Box* variations, but this requires more precision timing than a typical group of six will give you!

Kill Screen

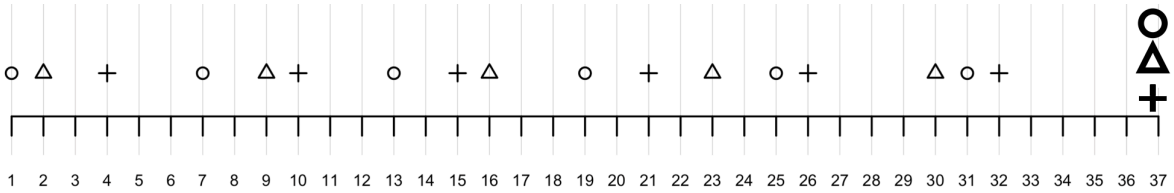
A non-repeatable pattern!

This pattern has a *lifespan* rather than a *cycle*! It is designed to crash out at exactly 37 1-count beats.





The brainchild of Mark Hayward and with specific passing sequence by Prof Mike, the *Kill Screen* pattern mimics the feature in older arcade and 80s console games that renders a level impossible to complete due to bugs or integer overflows. In this juggling rendition, pair (**A**,**D**) pass on 6-count, pair (**C**,**F**) pass on 7-count, starting on beat 2, and pair (**B**,**E**) pass on alternating 6-count/5-count, starting on beat 4.



The consequence of this particular passing sequence is that clubs avoid one another until the critical beat 37, at which point all passes collide in the middle to a tremendous cheer. It can be helpful to have a metronome to keep precise timing (remember, we like 127 bpm.) As Nathan Nard predicted theoretically and we confirmed empirically, if a common club color is in the 1st release position of all six jugglers, then that common club is also released into the collision on beat 37. Further, if passes are too precise they can actually all miss even on beat 37! One way to avoid this unhappy event is to have jugglers release more towards the centerline of their passing zone.

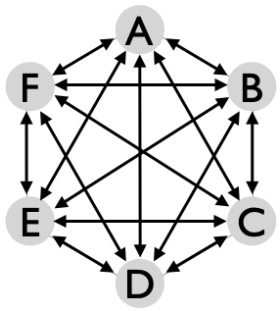
Metrics: J=5, pH=1, CJ=3, R=3, motion=0, spin=0, PBJ=0.18. Pattern born February 2024.

Feast (for 6)

Everyone feeds everyone else (clockwise-from-above version)

		Juggler

Pass sequentially around the circle, including yourself in the count. Any base count of choice works; 3 and 4 count are pleasant; 2-count requires some good passers and good timing. 1-count is a dream!



Beat	A	B	C	D	E	F
1	B	A	F	E	D	C
2	C		A	F		D
3	D	C	B	A	F	E
4	E	D		B	A	
5	F	E	D	C	B	A
6		F	E		C	B

MAJ have been juggling feasts since at least 1992, though their origin is unknown to us! *Feast* on a friendly count is a great first pattern when a group of non-beginner passers starts a session.

Metrics: J=6, CJ=3, pH=1, OJ=5, R=1, cycle length=6 on base count; PBJ=$[5/6]/[base]$, e.g., 0.28 on 3-count; 0.21 on a 4-count.. Some spin is needed to square up with your counterpart around the circle; considering the interior angles of a hexagon, everyone spins $4\pi/3$ rad = 240° per cycle. As presented the motion=0, but with 6 or more jugglers, we start to see an *amoeba* effect in which each juggler moves through a small circle in order to get closer to their passing partner as they proceed around the pattern. This motion reduces the variation in throw lengths. Let's say motion = ϵ for a small number ϵ, probably less than 1 pu in this 6-person version.

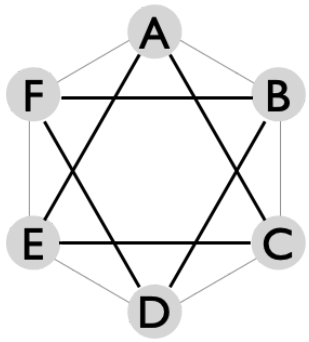
Variations include switching the direction, or, on 3-count:

Picnic

Start by practicing a 3-count feast. In the *picnic*, you complete one cycle throwing right-right or left-left crossing passes, then you switch on the next cycle (after doing your self beat) to do a round of straight passes (right-left or left-right). To set up, note that each pair of passers must include one *straighter* and one *crosser*. Picnic leads to all kinds of interesting Jim's jams, including long sequences of only left throws or only right throws. Just remember, pass back the club you catch! [*Homework: figure out the cycle length!]*

Star of David

A classic 6-person variation of the synchronized 4-person box



	Juggler					
Beat	**A**	**B**	**C**	**D**	**E**	**F**
Rights	**E**	**F**	**A**	**B**	**C**	**D**
Lefts	**C**	**D**	**E**	**F**	**A**	**B**

The *outsides* triangles version is shown; switch rights and lefts for the *insides* triangles version.

Any beat of choice works. Throws are simultaneous. In the *outsides* triangles version, right throws go to the person two spots to your right; left throws to the one two spots to the left. Feels like the 5-person *Star,* but club interference is a bit more likely; if trouble, try an *insides* triangles version.

Metrics: J=6, CJ=3, pH=1, R=1, PBJ=1/[base-count]. On an even base, OJ=1 and cycle length=1; on an odd base (e.g., 3-count), OJ=2 and cycle length=2. Motion=0, and spin=0 on even counts; but jugglers need some rotation on odd counts to sweep between their counterparts. We estimate spin = $2\pi/6$ rad = $120°$ for full cycle.

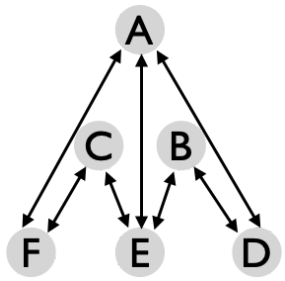
Collisions are certainly a possibility with this classic pattern, and so the need for precise timing makes it a bit difficult for beginning passers, even on a slow beat. (You're also not passing to the same person you receive from.) When well timed, your right throws will pass just behind the right throw of the juggler on your right, so keep an eye out! By the same token, any left throws pass immediately behind the left throw of the juggler on your left.

Feasts of David

Like the *Feast-Star* combinations for 5 jugglers, there's a lot of fun to be had by alternating *Feast* and then *Star of David* passes!

Overlapping Y's

An old MAJ pattern from the archive



	Juggler					
Beat	**A**	**B**	**C**	**D**	**E**	**F**
1	**E**	**D**		**A**	**B**	
2	**E**		**F**		**C**	**A**
3						
4	**D**	**E**		**B**	**A**	
5	**F**		**E**		**A**	**C**
6						

1-count base above for standard version

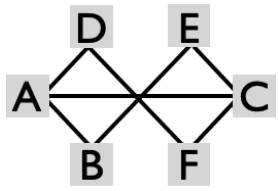
Recall the 4-person *Y* pattern, and imagine 2 copies where **A** and **E** are in both, being on one side of a *Y* with **C** and **F** and on the other side of a *Y* with **B** and **D**. Each *Y* is doing its own thing, and the overlap works as long as the **AE** passes are at different points in the two *Y*'s. The standard is a bit tough; each *Y* is on 3-count, so **A** and **E** end up being both on 3/2 count with some tricky throws. Right-handed 4-count Y's are easier, and then the overlap is lopsided (e.g., **A** receives from **D** but never passes to **D** in that version.)

Goofy variation: Like in the 4-person *Goofy Y,* **B** and **C** can be out of phase with the others, releasing lefts when everyone else is releasing rights. All the passes to and from **B** and **C** have to cross (right to left, etc), but owing to the positioning it all works out nicely!

Metrics: J=6, CJ=3, pH=1, R=2, OJ=2 ⅓, PBJ=4/9, cycle=6 1-counts, motion=0. Considering the swept angles for the ordinary *Y* and some distortion to fit everybody, the pattern average spin is about $\pi/3$ rad = 60°.

Overlapping feeds

An old pattern of unknown origin!

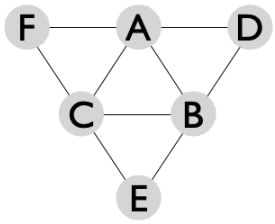


	Juggler					
Beat	**A**	**B**	**C**	**D**	**E**	**F**
1	**B**	**A**	**E**	**F**	**C**	**D**
2	**C**		**A**			
3	**D**	**E**	**F**	**A**	**B**	**C**
4	**C**		**A**			

Metrics: J=6, CJ=3, pH=1, R=2, cycle=4 on 2-count, OJ=2 ⅓, PBJ=⅓, motion=0, spin $\simeq \pi$ rad per cycle.

We have two 1 v 3 feeds that share jugglers. **A** feeds (**B**,**C**,**D**) and **C**, likewise feeds (**E**,**A**,**F**), presented here as windshield wiper feeds. The long middle passes make this one a bit challenging for **A** and **C**. Shortening the distance doesn't help too much owing to passing angles. We also add passes between the ends to keep them engaged! Standard is 2-count right-handed, but other timing works too.

Mel's Big Woody



Right-handed
2-count base

	Juggler					
Beat	**A**	**B**	**C**	**D**	**E**	**F**
1	**D**	**E**	**F**	**A**	**B**	**C**
2	**B**	**C**	**A**			
3	**C**	**A**	**B**			
4	**F**	**D**	**E**	**B**	**C**	**A**
5	**C**	**A**	**B**			
6	**B**	**C**	**A**			

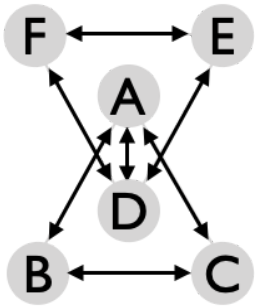
The three corners **F**, **D**, **E** have an easy 6-count to two interior feeders. These interior feeders **A**, **B**, and **C** are doing a 2-count sweep feed of 4 spots, and have a fair bit of rotation to get through each cycle. Note that the middle two beats of the sweep each forms a triangle.

Metrics: J=6, CH-3, R=2, pH=1, OJ=3, cycle = 6 on 2 count, PBJ=⅓, motion=0, spin=full turn per interior and less for the corners, with average $4\pi/3$ rad.

It takes a lot of skill on the inside triangle; a 3 or 4-count base is simpler but a bit slow on the corners! The corners have plenty of time; one variation is for them to pass triples to each other while the inside triangle is active, for instance on the same line as your last pass. E.g., **E** passes a triple to **F** after passing a single to **C**.

Argyle Socks

Closer quarters with both hands passing



Pattern by MAJ

	Juggler					
Beat	**A**	**B**	**C**	**D**	**E**	**F**
1	**B**	**C**	**A**			
2	**D**	C	B	**A**	F	E
3				**E**	**F**	**D**
4	**C**	**A**	**B**			
5	**D**	C	B	**A**	F	E
6				**F**	**D**	**E**

1-count base (both hands counting)

Standard right-handed start.

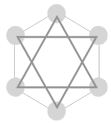
In the *basic version,* corners **F**, **E**, **B**, **C** do a 3-count outsides triangle. In both versions, **A** and **D** do an invigorating double 3-count, and in the *full version* all jugglers are on double 3-count (extra passes are indicated with grey font).

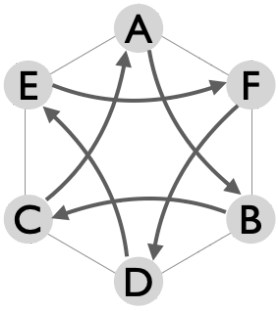
Tip: **A** and **D** keep your passes loopy!

Metrics: J=6, CJ=3, pH=1, R=2, OJ=2.67, PBJ=⅔ (*full)*, 4/9 (*basic*), motion=0. All jugglers swivel back and forth for spin= $2\pi/3$ rad = 120°.

Extra Club Stationary Six Person Patterns

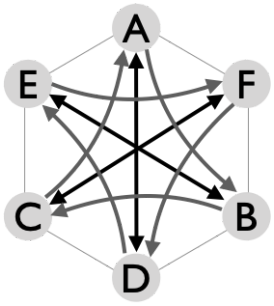
Overlapping extra club 4-count triangles





Start here!

Right beat		Juggler					
		A	B	C	D	E	F
1	0	d-**B**			d-**E**		
	1/3		d-**C**			d-**F**	
	2/3			d-**A**			d-**D**
2	0	**D**			**A**		
	1/3		**E**			**B**	
	2/3			**F**			**C**



Bubbler of David

This fun pattern pairing starts with equilateral two triangles (**A**,**B**,**C**) and (**D**,**E**,**F**) that overlap in space like a Star of David, and that are synchronized in time, with both doing a 1-extra club 4 count insides triangle. Recall the 3-person 10-club 4 count pattern, which each trio is doing, involves jugglers offset by ⅓ of a full right-hand beat, and passing insides double throws to the next person in the sequence, as shown above. As you master the synchronized triangles you realize that counterparts between the triangles are available to pass on the 2nd right beat of the cycle, and these we do by a crispy single pass straight across the pattern (grey shaded passes in table). The result we call the *Bubbler of David,* a name that merges the shape descriptor with the Wisconsin-y effervescent image of a water fountain. (Thanks Kat for that name!) It's a very doable pattern for good passers, but it requires good timing and patience!

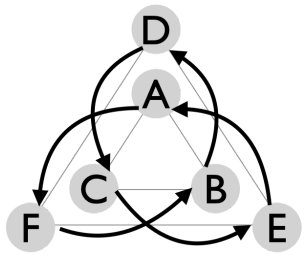
Metrics: J=6, CJ=3 ⅓, R=1, pH=2 (overlapping), 1 ½ (full Bubbler), PBJ = ¼ (overlapping), ½ (full Bubbler), motion=0. To square up with both recipients in the full Bubbler, each juggler swivels 120°, i.e., pattern spin=$2\pi/3$ rad. By our convention spin=0 in the overlapping version. The full Bubbler is also tricky because on the triangle throws you're not receiving from the person you pass to. OJ=1 in overlapping and 2 in full Bubbler. Pattern by MAJ.

Variation 7-Club 3-count Hex. For those seeking a challenge, try a Hex where each pair juggles a 3-count 7
club pattern in doubles. It works if the passes are precise. **I DOUBT IT!!**...maybe consider 21-club box in 2-count doubles or 4-count triples is more doable**

Trompe l'oeil

6-count version

Translated, "deceive the eye".



A starts with 4 clubs
All passes are doubles

1-count base		Juggler					
		A	**B**	**C**	**D**	**E**	**F**
A, B, C are out of phase from **D, E, F**	1	**F**					
	2						**B**
	3		**D**				
	4				**C**		
	5			**E**			
	6					**A**	

A delightful 1-extra club pattern for six stationary jugglers. All jugglers do 6-count rights passing to one other. It's helpful to vocalize your number as you pass. Inside positions are trickier owing to the bigger in-out angle and all the flying objects!

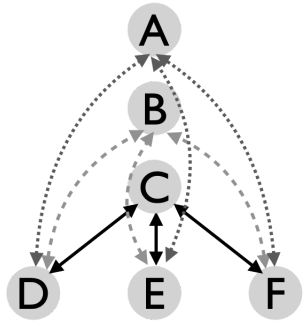
Metrics: J=6, CJ=3 ⅙, pH=2, R=1, cycle=6 on 1-count, PBJ=⅙, motion=0, spin=0 (recall our convention on spin refers to the angle swept out in passes over a cycle; since everyone has one recipient, we get spin=0 here, even though the inside jugglers have a big angle between their incoming and outgoing pass) . Pattern by Matt Mella. Matt learned a no-extra club version of the pattern with the same-to-from passing sequence, on any count of your choice. Noting the passing sequence, Peter Kaseman pointed out that this 1-extra club pattern is a stationary *Extra Club Double Weave*! **EXPLAIN THE 4-COUNT VERSION

Variation: 2-extra clubs. Same who-passes-to-whom, but now **D** and **A** start in phase with 4 each. Other jugglers are also in phase, passes are on 3-count, and are all right-right or left-left crossing throws. Then CJ=3 ⅓, PBJ=⅓, but other metrics are the same as above.

Theater-style Popcorn

A 3 extra club gem

A, **B**, and **C**, are each doing typewriter feeds of the same line, respectively in triples, doubles, and singles. Returns are popcorn: triple-double-single.



Right beat		Juggler					
		A	**B**	**C**	**D**	**E**	**F**
1-count base	1	t-**D**		**E**	t-**A**	**C**	d-**B**
	2		d-**E**				
	3	t-**E**		**F**	d-**B**	t-**A**	**C**
	4		d-**F**				
	5	t-**F**		**D**	**C**	d-**B**	t-**A**
	6		d-**D**				

*check, I might guess that D starts with C?**

All throws are right handed singles, doubles, or triples. **A**, **D**, and **F** start with 4. **B** is out of sync, starting with a left release on beat 1; everyone else starts with a right release.

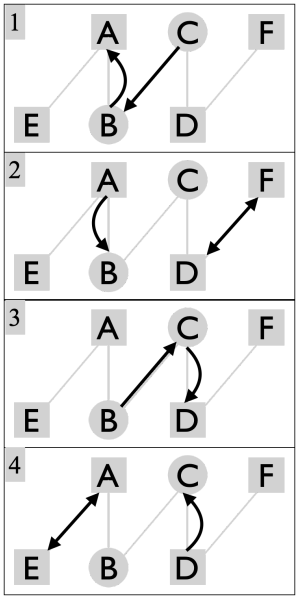
Metrics: J=6, CJ=3 ½, pH=2, OJ=3, PBJ=½, cycle=6 on 1-count, motion=0, spin=; R=4, with only **C** having a relatively easy role.

C needs to make crispy singles! Pattern by Peter Kaseman.

Graham/Claude's N+ with 2 caps

Double-single 2-counts

Graham Pasch got Claude's help with the N+ pattern, and then Mike added two 4-count singlers on the edge to cap it off!



		Juggler					
		A	**B**	**C**	**D**	**E**	**F**
1-count base	1		d-**A**	**B**			
	2	d-**B**			**F**		**D**
	3		**C**	d-**D**			
	4	**E**			d-**C**	**A**	

From October, 2026. It's a late bloomer but a very comfortable 1-extra-club pattern, with one double release on every beat. It's a rare case of out of phase jugglers in the same line.

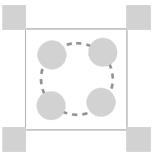
All passes are straight and right-handed. Central jugglers **A, B, C,** and **D** all do the same alternating single-double 2-count, while *caps* **E** and **F** do a basic 4-count. **B** and **C** (circles in figure) are in phase with each other and out of phase with the other four (squares). So **B** and **C** start with right-hand passes, while everyone else starts with a left self. Juggler **C** starts with 4 clubs; others start with 3 each. Loopy doubles and crispy singles make for good passing. [see 10/25/26 MCS video]

Metrics: J=6, CJ=3 ⅛ (1 extra), cycle=4 on 1-count, pH=1.4, OJ= 1 ⅔, PBJ = 5/12 = 0.42, motion=$4\pi/3$ rad, R=2.

Moving Six Person Patterns

Ball Bearing

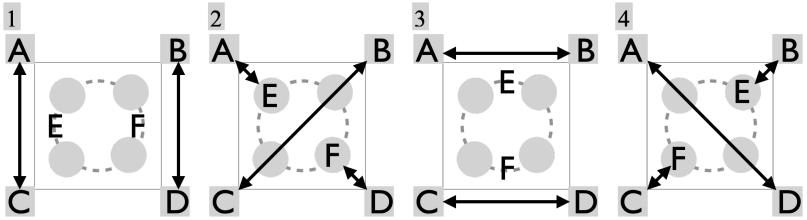
Here's a nifty extension of the 4-person *Feast*.



	Beat, 3-count base							
Juggler	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**
A	**C**	**E**	**B**	**D**	**C**	**F**	**B**	**D**
B	**D**	**C**	**A**	**E**	**D**	**C**	**A**	**F**
C	**A**	**B**	**D**	**F**	**A**	**B**	**D**	**E**
D	**B**	**F**	**C**	**A**	**B**	**E**	**C**	**A**
E		**A**		**B**		**D**		**C**
F		**D**		**C**		**A**		**B**

A, **B**, **C**, and **D** are doing a *Feast* (in 3 or 4-count), but the self throw is replaced with a pass to one of the movers in the middle. **E** and **F** are passing 6 or 8-count. On the 3-count base it's a bit easier to start lefts so that the long throws are right-handed.

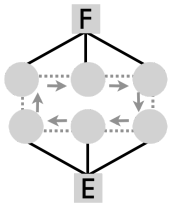
Metrics: J=6, CJ=3, pH=1, R=2, OJ=2 ⅔ , PBJ=5/18=0.28 (3-count base), cycle=8 beats on base count (3 or 4 count work); motion $\simeq 1$ pu (assume the circle for **E**, **F** has diameter about 1 pu). spin $\simeq 4\pi/3$ rad.



An old pattern by Mike Newton

Benzene Ring (for 6)

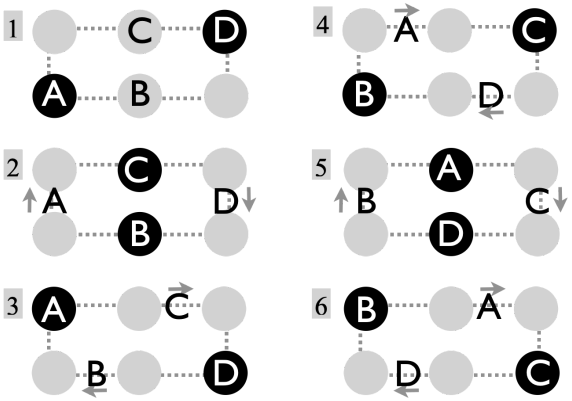
An MAJ standard



E and **F** each feed 3 positions: left, middle, right, left, middle, right, etc, while movers cycle through six positions.

	Beat, 2-count base								
Juggler	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**
A	**E**		**F**		**F**		**F**		**E**
B		**E**		**E**		**F**		**F**	
C		**F**		**F**		**E**		**E**	
D	**F**		**E**		**E**		**E**		**F**
E	**A**	**B**	**D**	**B**	**D**	**C**	**D**	**C**	**A**
F	**D**	**C**	**A**	**C**	**A**	**B**	**A**	**B**	**D**

Table shows ¾ of full cycle (12 2-counts)



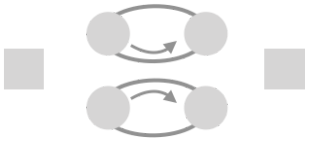
Feeders **E** and **F** pass simultaneous typewriter feeds with jugglers in the center, indicated with black circles. Movers shift to the next spot in the sequence after each pass. (½ cycle shown.)

Tip: Feeders pass deep throws to the position on the left, to help each mover switch sides.

Metrics: J=6, CJ=3, pH=1, R=2, OJ=, PBJ=, cycle=12 on 2-count base; motion ≃2 pu and spin ≃ 2π rad (1 full turn per mover and about the same per cycle for feeders). An old pattern by Mike Newton.

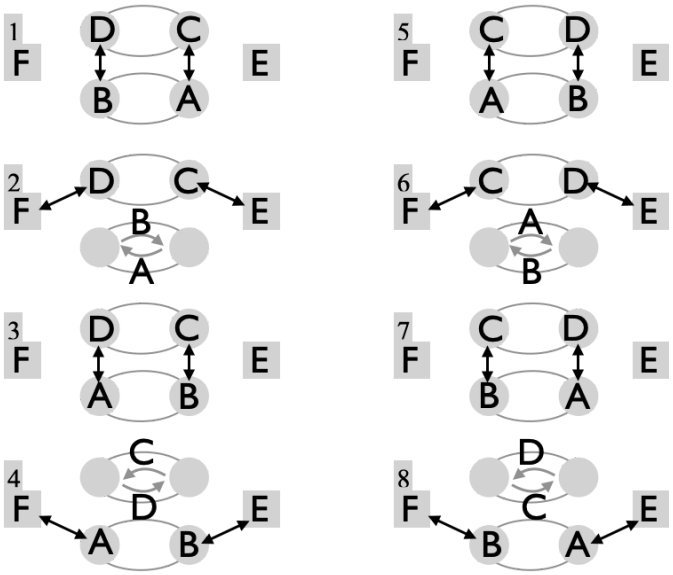
Two-handled eggbeater

A good pattern for beginning movers



	Beat, 4-count base							
Juggler	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**
A	**C**		**D**	**F**	**C**		**D**	**E**
B	**D**		**C**	**E**	**D**		**C**	**F**
C	**A**	**E**	**B**		**A**	**F**	**B**	
D	**B**	**F**	**A**		**B**	**E**	**A**	
E		**C**		**B**		**D**		**A**
F		**D**		**A**		**C**		**B**

Feeders **E** and **F** are handles. **(A,B)** is one beater and **(C,D)** is another.

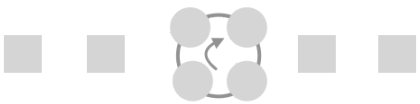


Derived by adding an extra juggler to the 5-person *eggbeater,* though it's hard to imagine what a real 2-handled eggbeater would look like! There's no reason why the pair switches can't either go in the opposite direction, as long as jugglers agree. Pattern by Mike Newton.

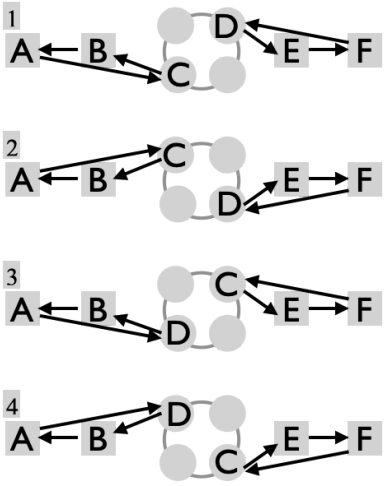
Metrics: J=6, CJ=3, pH=1, R=2, cycle=8 beats on 4-count base, OJ=4, PBJ=⅛, motion = 1 ⅓ pu; considering the swiveling back and forth the average spin is ≃ π rad per cycle.

Two-seated unicycle

Backdrops, spinning and both-hands passing!



Beat, 3 or 5-count base				
Juggler	**1**	**2**	**3**	**4**
A	C	C	D	D
B	A	A	A	A
C	B	B	E	E
D	E	E	B	B
E	F	F	F	F
F	D	D	C	C



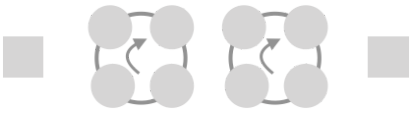
B and **E** are posts facing in towards the wheel (**C**,**D**). They drop back to seats **A** and **F**, and receive passes from the wheel, which turns clockwise from above.

Metrics: J=6, CJ=3, pH=1, R=3, OJ=1 ⅔, cycle is 4 beats on 3 or 5 count base. PBJ = 1/base. Motion is 2 pu for the movers, giving a pattern average ⅔ pu. **C** and **D** have substantial spin (a full turn per cycle). Pattern spin $\simeq 5\pi/6$ rad. Get started with the 4-person *unicycle*. Pattern by MAJ.

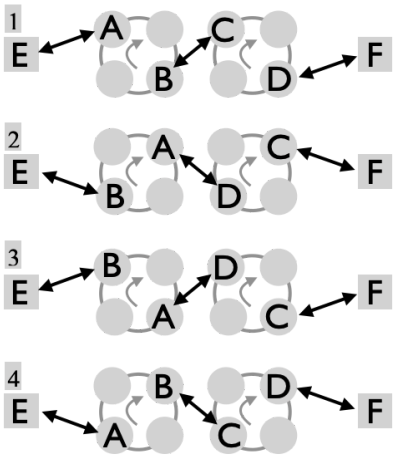
Razor

1-2-2-1

Fun with 4 good movers and 2 feeders.



	Beat, Base count			
Juggler	**1**	**2**	**3**	**4**
A	E	D	D	E
B	C	E	E	C
C	B	F	F	B
D	F	A	A	F
E	A	B	B	A
F	D	C	C	D



The four movers **A**, **B**, **C**, and **D** are busy, spinning a ½ turn after every two throws and moving quickly to cover a full circle in 4 passing beats. The pairing has everyone passing twice in a row to the same partner.

Tip: Make the second throw of each pair deep enough to support the direction changes.

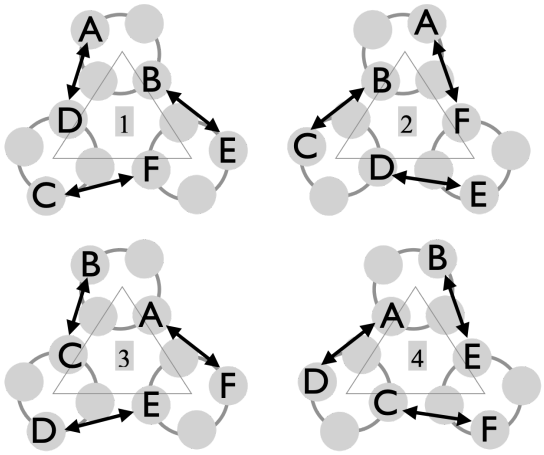
Pattern by Peter Kaseman.

Metrics: J=6, CJ=3, pH=1, R=2, OJ=2, cycle is 4 beats on the base count. 3-count is the high standard, but you'll want to start on slower counts. Average motion = 1 ⅓ pu; spin is 2π rad per mover, and very slight per feeder, for an average in excess of $4\pi/3$ rad.

Electric Razor

An MAJ standard

	Beat, 3-count base			
Juggler	**1**	**2**	**3**	**4**
A	**D**	**F**	**F**	**D**
B	**E**	**C**	**C**	**E**
C	**F**	**B**	**B**	**F**
D	**A**	**E**	**E**	**A**
E	**B**	**D**	**D**	**B**
F	**C**	**A**	**A**	**C**



Jugglers pair up. In each of the three pairs, jugglers cycle around each other while passing twice in a row to one member of each other loop, with all passes on 3-count.

Tip: Move quickly and juggle slowly.

Pattern by Peter Kaseman, summer 1998.

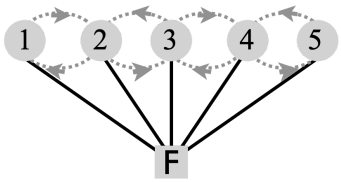
On December 1, 2022, we (Chan, Jeff, Mike, Mel, Nathan, and Nick) revisited the *Electric Razor* with colors, and on a comfortable 5-count base (rather than 3-count base). It takes 3 cycles to get back to the start. Good leads are important here, but they are deceptively tricky when you're moving fast enough for the whole pattern to work.

Metrics: J=6, CJ=3, pH=1, R=1, OJ=2,PBJ=⅓, cycle=4 beats on 3-count, motion ≃ 2 pu. Each juggler swivels approximately ½ turn through each cycle: spin ≃ π rad.

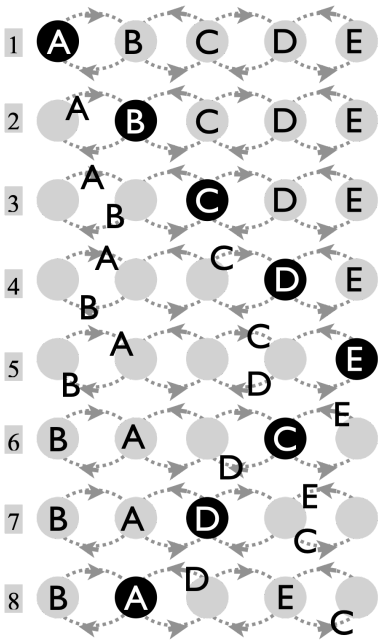
Peter Feeds 5

Aka: *5-person weave*; aka *24153*

MAJ have been juggling this since 1998, though its origin is unknown!



Juggler	Beat, 2-count base, ⅖ cycle															
	1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**13**	**14**	**15**	**16**
A	**F**							**F**			**F**			**F**		
B		**F**							**F**							**F**
C			**F**			**F**							**F**			
D				**F**			**F**			**F**						
E					**F**							**F**			**F**	
F	**A**	**B**	**C**	**D**	**E**	**C**	**D**	**A**	**B**	**D**	**A**	**E**	**C**	**A**	**E**	**B**



⅕ full cycle

This fun pattern requires straightforward movements but is mentally tough for beginners since the count changes as you move through the weave. The feeder (**F**) performs a windshield wiper 2-count feed of the 5 positions, vocalizing 1, 2, 3, 4, 5, 4, 3, 2, 1, etc. Movers advance through the spots.

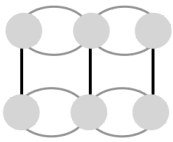
The local rule for each mover (**A-E)** is: pass to Peter (**F**) then move to the next spot. If you're in an odd-numbered spot, then you move back away from Peter and *with the grain* (i.e. in the same direction as Peter's sweep); if you're in an even spot then you move forward and *against the grain*. You end up passing more frequently in the middle and less frequently on the ends. The shortest gap is 6-count; patience is a virtue as you move towards the ends! The figure above shows one full sweep back and forth by the feeder (8 2-counts). A full cycle takes 5 full back and forth sweeps (40 2-counts). It's entertaining to recognize that the action of the pattern permutes the 5 movers, with the specific permutation $\sigma = (24153)$ associated with one full back-and-forth sweep. Two states traversed by the action of this permutation as Peter feeds the pattern are highlighted in the table (bottom row, grey shading).

Metrics: J=6, CJ=3, pH=1, R=2, OJ=1 ⅔,, PBJ is 1/10 for each mover and ½ for **F**, giving a pattern average 1/6, cycle=40 on 2-count base, and average motion=16 pu. The feeder sweeps back and forth 5 times for a full cycle; the precise spin depends on how far **F** sets up from the line, but if **(A,E,F)** start as an equilateral triangle, then **F** spins $10\pi/3$ rad and each mover must spin about $2\pi/3$ rad, for a pattern average $\simeq \pi$ rad per cycle.

As you've seen, there is a mind boggling variety of possible weaves, which can be paired up in many ways to make a host of 6-person variations. Here are some we like.

Standard Double Weave

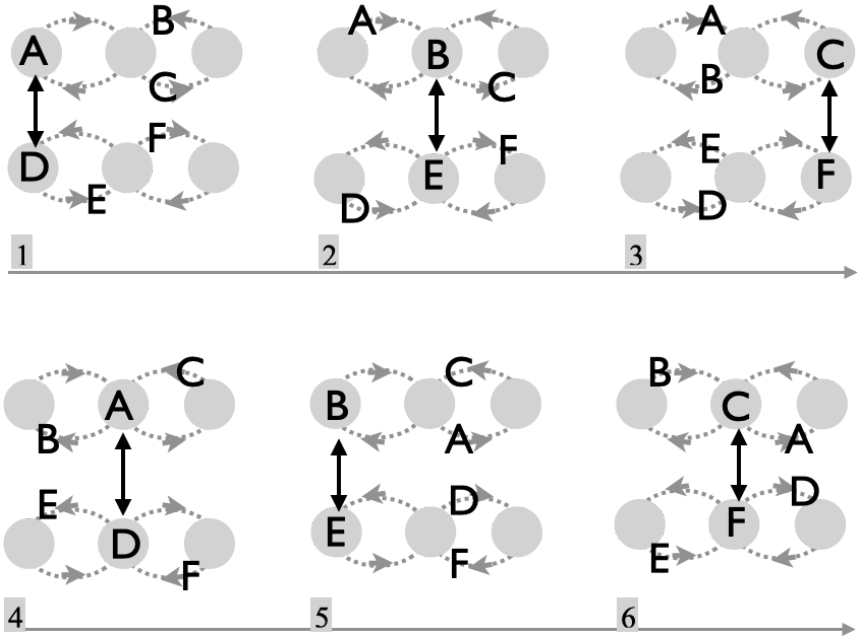
A building block for movers



Jugglers move in and out of passing points, the grey circles.

	Beat, 2-count base		
Juggler	**1**	**2**	**3**
A	**D**		
B		**E**	
C			**F**
D	**A**		
E		**B**	
F			**C**

Two sets of three weavers each set up facing each other. ½ cycle shown right.



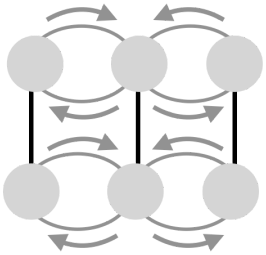
The *Standard Double Weave* has each juggler moving forward in the middle, and back on either end. Move to the next position after each pass. This extends the 4-person *Standard Weave* by replacing the 2-count feeder with three 6-count weavers. To compensate for the motion, end passes ought to be deep and middle passes ought to be short, while all aiming for the same height to keep proper time. As in the 4-person weave, there's a tendency for beginner weavers to make a very shallow figure 8. Emphasize those back-on-the-end moves if that happens. And don't forget: lead your partner, juggle slow, and move fast! Coupling this pattern with two cycles of a *Feast* is a favorite club-passing-routine combo pattern, especially with a nice synchronized triple-up at the end (*Double-Weave-Feast*).

Metrics: J=6, CJ=3, pH=1, R=1, OJ=1, PBJ=⅛, cycle=12 on 2-count base; local passing cycle=3. Motion=4 pu and spin=0, since you're always squared up with your passing partner.

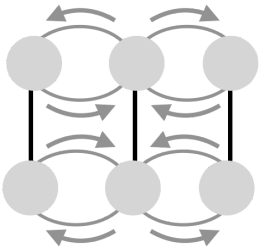
Inverted and Dresser Drawer Double Weaves

Vary the motion.

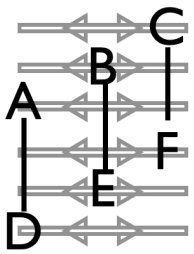
Refer to Standard Double Weave for passing sequence.



One weave inverted



Both weaves inverted



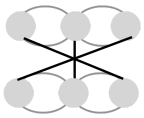
Weavers slide back and forth

Two notable variations of the *Standard Double Weave*, which share passing rules and metrics with the standard, have weavers on one (or both) sides moving differently than the standard back on the end, forward through the middle. An inverted weave has, *back in the middle forward on the ends*. This one can be a bit easier than the standard because passing pairs stay equidistant throughout the pattern when one weave is inverted and one is standard. But it requires the inverted weavers to take care coming off the ends so as not to move prematurely into the middle and get hit by a middle pass! It also seems to be trickier to accelerate backwards in the middle compared to an end, as there's less room to work in. The other variation has the weavers actually sliding back and forth in the dresser-drawer format (see 4-person *DresserDrawer Weave*).

Metrics: J=6, CJ=3, pH=1, R=1, OJ=1, PBJ=⅛, cycle=12 on 2-count base, local passing cycle=3, motion=4pu, and about $2\sqrt{2} \simeq 3.8$pu for the dresser drawers.

Cross Corner Double Weaves

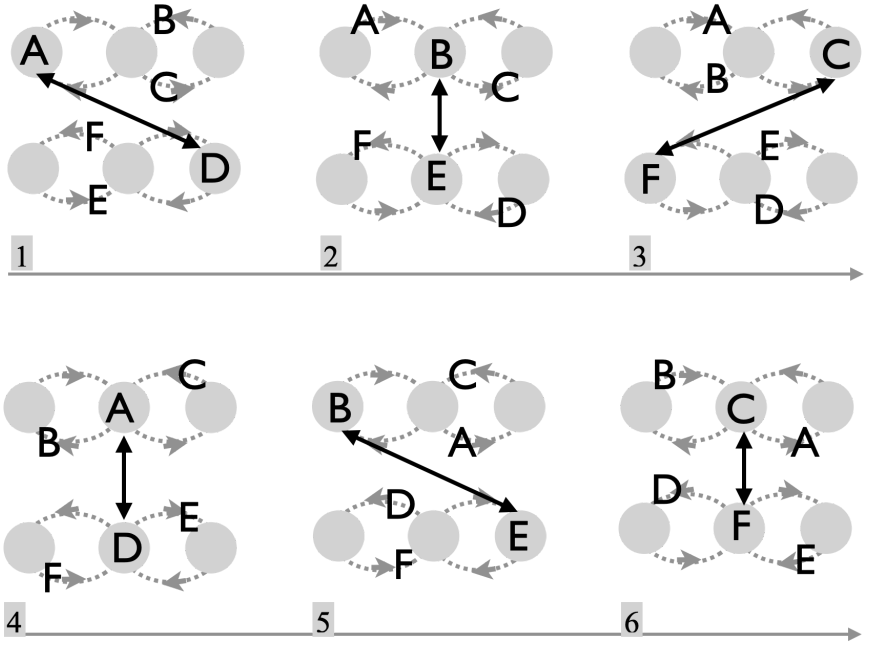
Master the standard, then take it up a notch.



Jugglers move in and out of passing points, the grey circles.

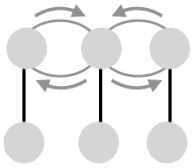
	Beat, 2-count base		
Juggler	**1**	**2**	**3**
A	**D**		
B		**E**	
C			**F**
D	**A**		
E		**B**	
F			**C**

Two sets of three weavers as above. ½ cycle shown right.



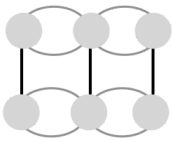
The passing sequence here is identical to the *Standard Double Weave*, but instead of lining up opposite your partner, each end weaver lines up on the opposite side of the pattern from their passing partner. It's as if one of the weaves (e.g., **D, E, F**) is a mirror image of what it was in the standard pattern. This one is a notch more difficult than the standard, in part owing to extra rotation (spin) as you square up to your passing partner, and also because those cross-corner passes are long and have to be quite precise to avoid hitting the movers heading through the middle! Also called the *Kitty-Korner Weave*

Metrics: J=6, CJ=3, pH=1, R=1, OJ=1, PBJ=⅙, cycle=12 on 2-count base, local passing cycle=3, motion=4pu and spin is about $\pi/2$ rad per cycle.

Unweave		One stale weave
Less notable double-weave variations amount to freezing three of the movers in place, and having just one set of three do the weaving. This is a great pattern when your group includes a few beginning passers who have not yet mastered moving and juggling. In the standard version, the weavers are on 6-count and the stationary line has a 4-counter in the middle and two 8-counters on the outsides. Other count variations are fun too.		
Metrics: J=6, CJ=3, pH=1, OJ=3, R=3, PBJ=⅛ , cycle=12 on 2-count, motion=2pu, spin=0.		

Stick on the End Double Weave

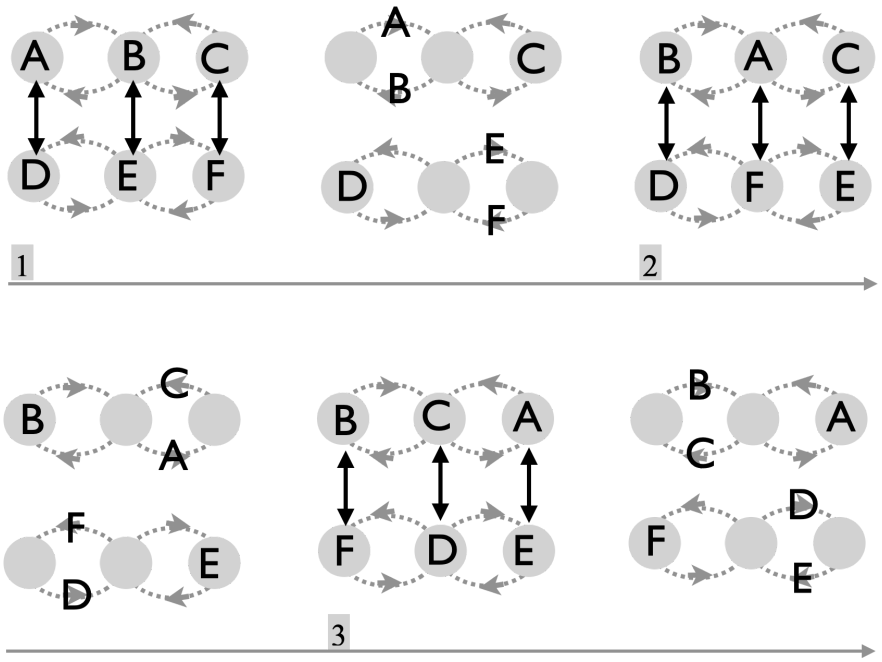
Synchronicity.



Move like other weaves, but pass twice on each end.

	Beat, 6-count base		
Juggler	**1**	**2**	**3**
A	**D**	**F**	**E**
B	**E**	**D**	**F**
C	**F**	**E**	**D**
D	**A**	**B**	**C**
E	**B**	**C**	**A**
F	**C**	**A**	**B**

Two sets of three weavers. ½ cycle shown right, with intermediate moves.



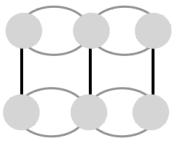
Prof Mike remembers working on this pattern in the Tripp Commons with Matt Henry, Mark Hayward and other MAJ as early as 1993. It remains one of our favorite double weaves. You don't have a single passing partner, as in other double weaves, but instead you pass to all others as you move the figure 8. Key to making it work is the *stick-on-the-end f*eature, which has you pass twice at each end position. In the set up above, **C** and **D** start as stickers. Owing to the synchrony, there's also no restriction to the typical 6-count passing rate. Try 5-count or even 4-count if you're feeling your oats!

Metrics: J=6, CJ=3, pH=1, R=1, OJ=3, PBJ=⅛ (or ⅕, or ¼), cycle=6 on 6-count base; local passing cycle=3, motion=4pu and spin=0.

Note: If each side moves on the edges and middle of a rectangle, instead of weaving, we have the essence of Han's Delight. \[Euro '98'\].

Mondo Double Weave

A new member of the double-weave family.

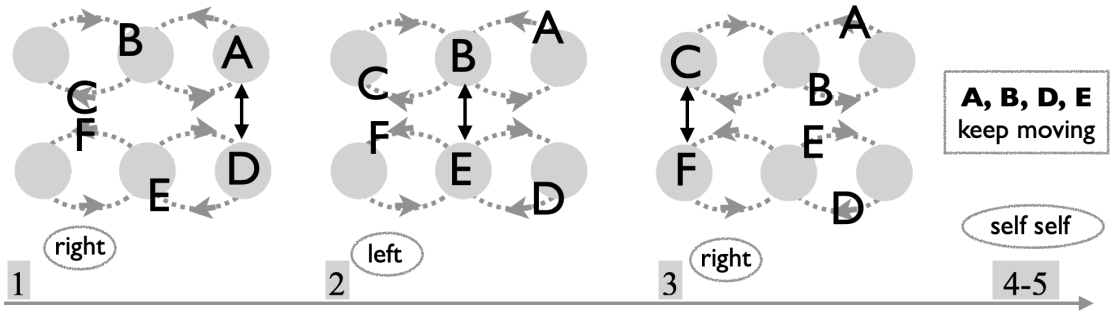


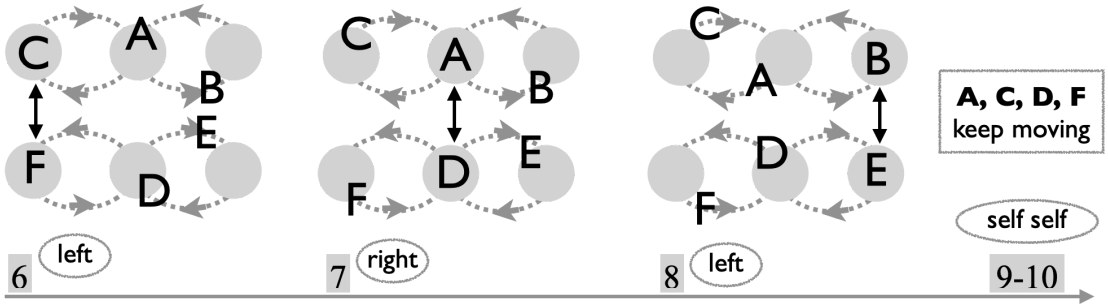
	Beat, 1-count base (1/2 cycle)														
Juggler	**1**	**2**	**3**	**4/5**		**6**	**7**	**8**	**9/10**		**11**	**12**	**13**	**14/15**	
A	D						D						D		
B		E						E			E				
C			F			F						F			
D	A						A						A		
E		B						B			B				
F			C			C						C			

Everyone passes on two 6-count gaps followed by a 3-count gap.

The pattern has a curious stick-on-the-end feature but also an asynchronous feature, in contrast to the ordinary stick-on-the-end double weave. Each weaver also has a single partner, like the standard double weave. So it's a true hybrid, discovered just recently (Feb 2026) when visiting Mondo and extending their 4-person `6-count` weave to a double weave format. Thanks Tom!

In the steady state, each pair passes three rights in a row on 6-count, followed by a 3-count stick on the end, followed by three lefts in a row on 6-count going back the other way.





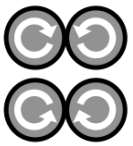
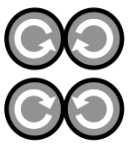
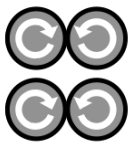
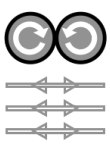
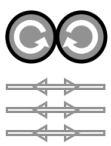
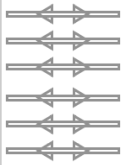
Metrics: J=6, CJ=3, pH=1, OJ=1, R=1, PBJ=⅛, cycle length=30 on 1-count base, motion=4, spin=0.

Combinatorial Double Weaves

Explore the multitude.

Here's a fun game to play when you have six good movers. Try all 18 variations.

Even without cross-corners or switch-sides, we get 18 different double weaves by combining motion and pass variants.

		Type of movement					
		standard standard	invert invert	standard invert	standard dresser	invert dresser	dresser dresser
							
Type of 6-count passing	standard	✓		✓			✓
	stick	✓	**Work through all 18 with 5 friends!**				⚠
	mondo	✓		✓			⚠

Each of the two weaves can be either standard (i.e., back on the end, forward through the middle), inverted (back in the middle, forward on the ends) or dresser drawer (side-to-side). That gives 6 possible movement types (columns). Independently, the pair of weaves can select one of three different passing types (rows). For instance, standard refers to the asynchronous 6-count deployed in the *Standard Double Weave,* stick refers to the synchronous passing in the *Stick-on-the-End Double Weave,* and mondo refers to the Minnesota 6-count. Check marks are ones we do regularly, but it's worth the effort to try all of them. Warning signs indicate difficulty owing to passing lengths!

Metrics: Common to all are: J=6, CJ=3, pH=1, spin=0. OJ depends on the type of passing, with OJ=1 (standard), 3 (stick, mondo). Likewise, PBJ=⅛ for standard and stick, and ⅛ for mondo. Motion is 4 pu in the first three columns and a bit less for anything with dresser moves.

Add even more combinatorial complexity by including the *inside-out weave* as one of the movement types!!

MORE FUN (from Aug 6, 2026)

Reconsider the Mondo-weave, and notice the weave spits out 3 passes every 5 beats. Two things we observe: (1) you can replace a single feeder with 3 stationary 5-count feeders, the middle one only passes to the middle, and each end feeder passes to the two sides of the weave. And (2), those 3 5-counters can start to move, in a 5-count standard weave…that's fast, but it's completely different from the opposite Mondo weave which sticks on the end. In other words, we can mix types (mondo, standard), as long as the standard is sped up to 5 count..very cool. More than that, the Mondo p-p-p-s-s base sequence can also be replaced by a p-p-s-p-s sequence, giving the same

average! The movers end up doing 2-count stick on the end, followed by 6-count off the end and 7-count out of the middle!

Switch Sides Double Weave

Switch sides at each end. For example, after A and D pass at the end, they switch positions so that A, E, and F are weaving, as are D, B, and C.

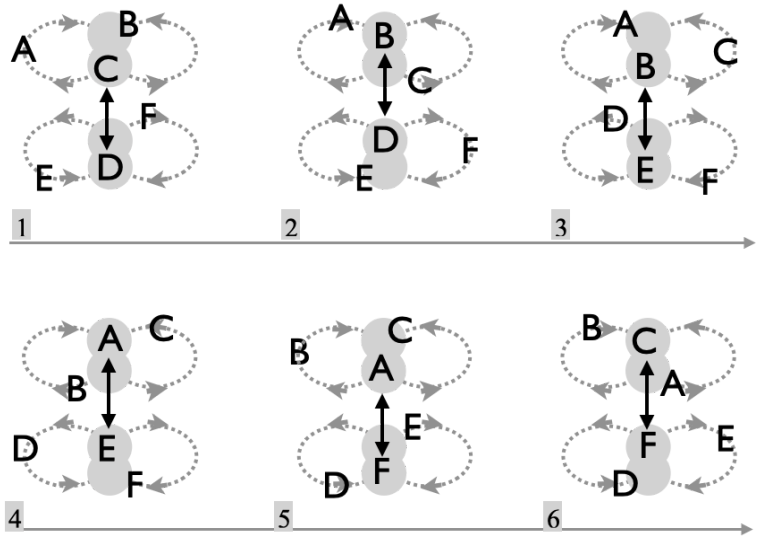
Double Gandini Weave

A double weave extension of the 4-person *Gandini Weave*

By Gandini Juggling Project and MAJ

	Beat, 2-count base					
Juggler	**1**	**2**	**3**	**4**	**5**	**6**
A				**E**	**F**	
B		**D**	**E**			
C	**D**					**F**
D	**C**	**B**				
E			**B**	**A**		
F					**A**	**C**

Standard version is right-handed 2-count. Passes only happen in the middle.



Each weaver passes twice in a row to two different jugglers, and then has 4 selves to move out, around, and back and into the middle. The next move out goes into the opposite lobe. You'll be surprised how fast the motion needs to be to meet the 2-count base, so try on a slower base at first. Don't forget to leave the middle in the opposite direction from the last mover. To start, it works well to stack jugglers **(C,B,A)** in one line with **C** in front facing another stack **(D,E,F)** with **D** in front. Again, the basic rule is pass twice then phantom on 4 beats while you move out of the way for the others. Lot's of timing variations fit that format, and the two sides may be in or out of sync in the directions they move out of the middle.

Metrics: J=6, CJ=3, pH=1, R=1, OJ=2, PBJ=⅛, cycle=12 on 2-count base, motion=4pu, spin=0.

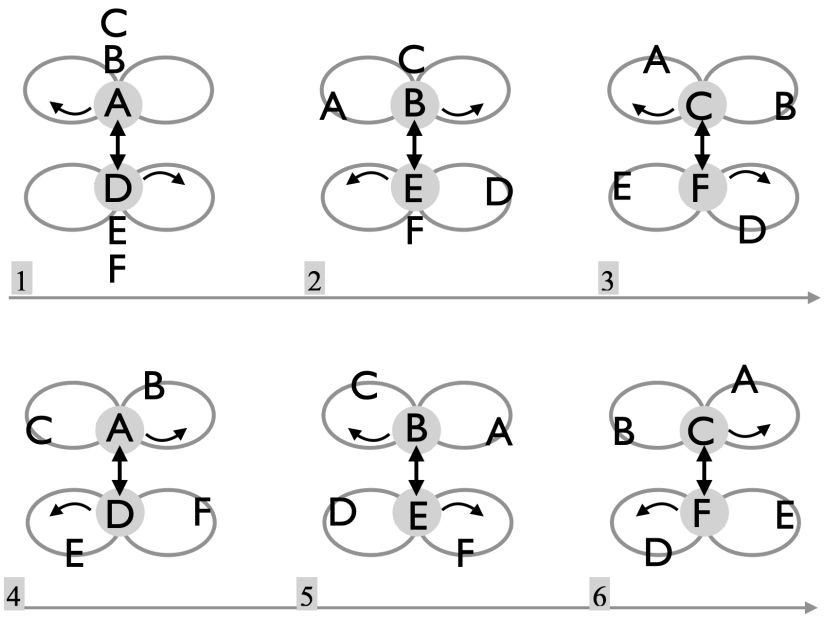
Double Gandini Site-Swap Weave

3v3

A surprisingly quick 9-count!

	Beat, 3-count base		
Juggler	**1**	**2**	**3**
A	**D**		
B		**E**	
C			**F**
D	**A**		
E		**B**	
F			**C**

Move right after passing right; move left after passing left.



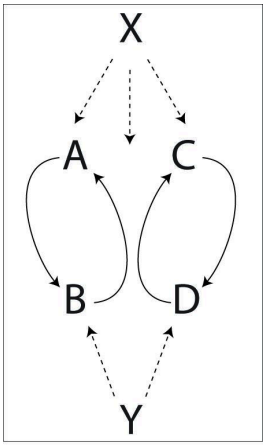
Metrics: J=6, CJ=3, pH=1, OJ=1, R=1, local cycle = 3; full cycle=6 on 3-count, PBJ=1/9 Motion through figure-8 as in *standard weave*, giving pattern motion=4pu per cycle. Since passes are always in the midline there's no spin at all (spin=0).

Site-swap variants: As with other Gandini Site Swaps, imagine each set of jugglers, (**A**,**B**,**C**) and (**D**,**E**,**F**) separately as three objects being juggled. A pass event corresponds to imaginary jugglers *throwing* each passer; odd beats correspond to right hand throws, both for the passers and for the two imaginary jugglers, and likewise for even beats and left-hand throws. Shown is the simplest pattern, where the two imaginary jugglers are each doing site swap: **3**. Any site swap should work, and they don't need to be the same on the two sides! We've explored more of the site-swap patterns with 5 movers and one feeder who takes the role of the imaginary juggler, but the 3 versus 3

Twister 4 Weave

Twister weaves can be done with any number of jugglers. Here's a six person version.

	Beat, 2-count base							
Juggler	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**
A	**X**			**Y**		**X**		
B		**X**			**X**			**Y**
C			**X**			**Y**		**X**
D		**Y**		**X**			**X**	
X	**A**	**B**	**C**	**D**	**B**	**A**	**D**	**C**
Y		**D**		**A**		**C**		**B**



Metrics: J=6, CJ=3, pH=1, R=3, OJ=2 ⅔ , PBJ=¼, cycle=8 on 2-count base, motion=1 ⅔ pu (assuming movers cover about 2pu each per cycle). Movers complete one full turn per cycle, and feeders perhaps a quarter turn, for a pattern average spin $3\pi/2$ rad.

Pattern by Luke Emery.

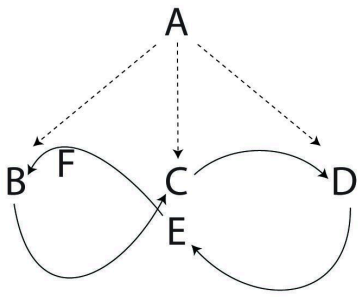
5 Weaves

These weaves have 1 feeder and 5 moving feedees

Both versions

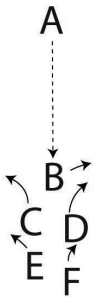
	Beat, 2-count base				
Juggler	**1**	**2**	**3**	**4**	**5**
A	**B**	**C**	**D**	**E**	**F**
B	**A**				
C		**A**			
D			**A**		
E				**A**	
F					**A**

Standard 5-Weave



Passes happen in the middle and on both ends.

Gandini 5 Weave



Passes happen in the middle.

Standard: After passing, move back if on the end and forward if in the middle. The start can be confusing to sort out, but set up as in the figure, with the first weaver who passes in the middle (**C**) going away from the first outside pass.

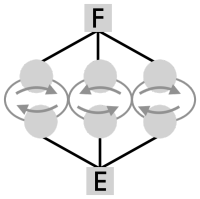
Gandini: Start with the weavers stacked in a line all facing the feeder. Then each weaver passes once and moves in the opposite direction of the preceding weaver. We learned about this family of weaves from Sean Gandini and the Gandini Juggling Project in 2000. Thanks!!

Metrics: J=6, CJ=3, pH=1, R=3, OJ=5/3, PBJ=1/6, full cycle=10 beats on 2-count base (Gandini), local cycle is 5. Loops are bit bigger than standard weave to accommodate all the movers, who each each cover about 5pu per cycle, for a pattern average motion about 4 ⅛ pu. There's no spin in the Gandini style version, but the standard version requires swiveling like in a *standard weave,* giving a pattern average spin **. *How many sweeps per cycle?*

Many of the timing and motion weave variations for 3 weavers carry over. We especially like the site-swap variations of the Gandini weave ***EXPLAIN*** prefer 3-count base

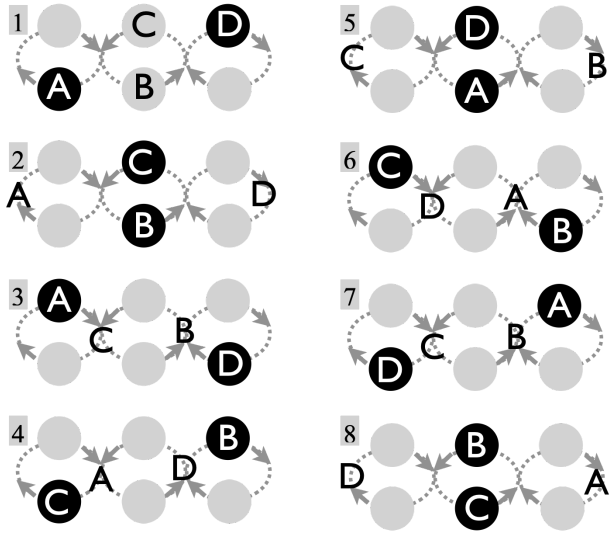
Spiral Weave

Peter and Luke have complained that this shouldn't be called a *Spiral*, but that's what we call it! It's sort of a weave, but there are three loops instead of two. And you pass at the faces of the loops instead of on the edges.



	Beat, 3-count base (⅔ cycle)							
Juggler	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**
A	**E**		**F**		**E**		**F**	
B		**E**		**F**		**E**		**F**
C		**F**		**E**		**F**		**E**
D	**F**		**E**		**F**		**E**	
E	**A**	**B**	**D**	**C**	**A**	**B**	**D**	**C**
F	**D**	**C**	**A**	**B**	**D**	**C**	**A**	**B**

Passes to the spiraling movers are indicated as solid black circles.



The standard version has weavers in 6-count (**A** and **D** do 6-count rights and **B** and **C** do 6-count lefts) and feeders in 3-count carriage return. Beginners could try feeders in 4-count and movers in 8-count. The *Fast Spiral,* which has feeders on 2-count and movers on 4-count, is a high-level pattern; movers switch sides on 4-count and feeders need to send just-right leads to make it work! Pattern by Mike Newton and the MAJ. The *Hyper Spiral* has weavers in 3-count and feeders on 3/2 count; but it's still a dream!

Metrics: J=6, CJ=3, pH=1, R=2, OJ= 2 ⅔ , PBJ=2/9 (3-count base), ⅓ (2-count base). Full cycle is 12 base beats. Movers cover about 6 pu per cycle, making the pattern average 4 pu. Movers also complete 4 full turns, and feeders swivel almost 2 full turns as they each perform 4 carriage returns per cycle, for a pattern average spin $\simeq 6\pi$ rad. (That's a spinny one!)

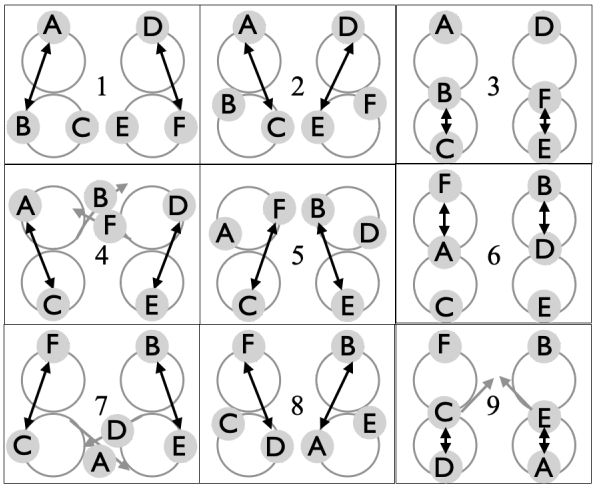
Interlocking Turbos

A classic MAJ pattern. [Pre-requisite is 3-person *Turbo*]

Our favorite among several ways to conjoin two *Turbos*.

	Beat, 2-count base, ½ cycle								
Juggler	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**
A	**B**	**C**		**C**	⤻	**F**	→	**B**	**E**
B	**A**	↷	**C**	→	**E**	**B**	**E**	**A**	
C	←	**A**	**B**	**A**	**F**		**F**	↷	**D**
D	**F**	**E**		**E**	⤾	**B**	←	**F**	**C**
E	→	**D**	**F**	**D**	**B**		**B**	⤻	**A**
F	**D**	↶	**E**	←	**C**	**A**	**C**	**D**	

Standard version is right-handed 2-count. Switch sides ↷; slide →. Feeder positions are shaded in the table.



Jugglers set up as in panel 1, with both first-feeders (**A**,**D**) facing the others. It's helpful to recognize a mirroring of positions (e.g., to simplify walkthrough). For instance **B** and **F** are mirrors and so are **C** and **E**. Mirrors have the same passing and motion sequence. The basic rule is to cross over into the other *Turbo* when it's your turn to go through the middle. To avoid body blows, jugglers need to decide who goes in front during this transition and who goes behind! In spite of the symmetry, the two *Turbos* are not exactly the same, since jugglers (**A**,**E**,**F**) do *inside* turns (counter-clockwise from above) and jugglers (**D**,**B**,**C**) do *outside* turns (clockwise from above), which are usually a bit trickier with right-hand passing. Hand and count variations are fun; e.g. 3-count is good for beginners

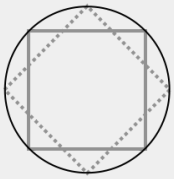
Warm start: Have (**A**,**B**,**C**) and separately (**D**,**E**,**F**) initiate ordinary 3-person *Turbo*s on either side of the room, running each for a full 18-beat cycle as they advance towards each other. When these separate patterns are side by side, they are then ready to interlock.
Disengage: This accompanies the warm start, and refers to separating the two *Turbos*, usually after ½ cycle interlocking or 3/2 of such a cycle. The whole business gives the impression of two 3-person patterns colliding, then moving through each other, and coming out the other side!

Metrics: J=6, CJ=3, pH=1, R=1, OJ=4, PBJ=⅓, cycle=18 on 2-count base, but passing sequence repeats after 9 beats. There's some extra lateral motion compared to a single *Turbo*, so motion is about 6pu per cycle and spin = 2π rad. Pattern by Luke Emery. It has been juggled often, including by the Mad-5 at the 2001 IJA Cascade of Stars show in Madison, 1999.

Back-to Back Turbos?

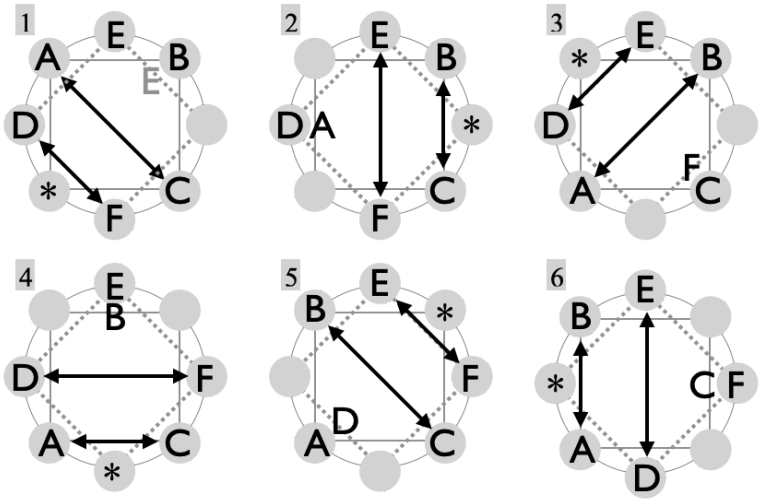
Windy Two Holer

A classic MAJ pattern born on a windy day on the Library Mall. Mel and Mike were there!



	Juggler					
Beat	**A**	**B**	**C**	**D**	**E**	**F**
1	**C**		**A**	**F**	⇑	**D**
2	⇑	**C**	**B**		**F**	**E**
3	**B**	**A**		**E**	**D**	⇑
4	**C**	⇑	**A**	**F**		**D**
5		**C**	**B**	⇑	**F**	**E**
6	**B**	**A**	⇑	**E**	**D**	
7	**C**		**A**	**F**	⇑	**D**
8	⇑	**C**	**B**		**F**	**E**
9	**B**	**A**		**E**	**D**	⇑
10	**C**	⇑	**A**	**F**		**D**
11		**C**	**B**	⇑	**F**	**E**
12	**B**	**A**	⇑	**E**	**D**	

⇑ indicates move



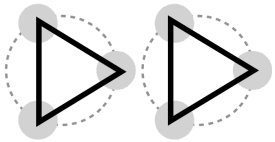
Despite its full cycle being the rather lengthy 24 passing beats, the *windy-two-holer* is not so hard to remember, since each juggler's task repeats every 6 passing beats. This is fun in 4-count, 3-count, and even 2-count for those not faint of heart. Six jugglers set up around a circle inscribed with eight spots corresponding to the corners of two inscribed squares. The jugglers cover six of the eight spots, leaving two *holes*, known affectionately as the *a-hole* (*) and the *number-2 hole*. The a-hole floats around while jugglers in sequence move into that spot.

Jugglers **A**, **B**, and **C** move along one square while **D**, **E**, and **F** move along the other. The two sets of jugglers don't interact (on purpose, anyway!). Each juggler only ever passes to the two others in his/her square, never to the other square. In the steady state, the local passing rule is, *pass one - pass other - self - pass one - pass other - move/self,* counting on the passing beats. At any passing beat one pair from each square is passing; collisions are avoided because the passing lanes are parallel. We start with **E** having already landed, but a steady-state start would have **E** on the move at the beginning (grey font, beat 1).

Metrics: J=6, CJ=3, pH=1, R=1, OJ=2 , PBJ= cycle=24 beats on base count (2, 3, or 4 count). Edges of each box are about 1pu (maybe a bit more), which makes motion per cycle 4 pu (same for all jugglers). Noting the swivelling back and forth per edge along 4 edges, the average spin per cycle is about 6π rad (3 full turns). (Note *spiral weave* has the same spin metric but over a pattern with ½ of the cycle of this one.)

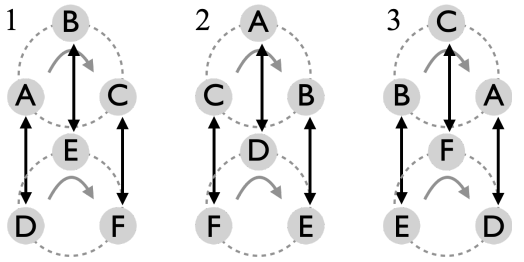
Fast Forward

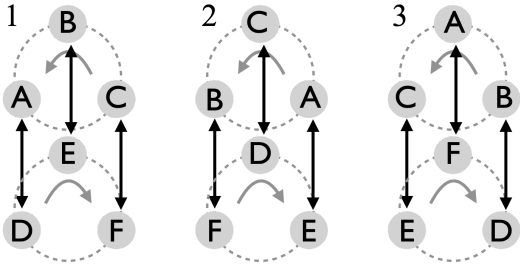
New-style non-weaving double weaves



Version 1	Beat, base count		
Juggler	1	2	3
A	D	D	D
B	E	E	E
C	F	F	F
D	A	A	A
E	B	B	B
F	C	C	C

Two sets of three movers as above.





Passing is synced up, so any feasible count works (e.g., 6, 5, or 4).

Version 2	Beat, base count		
Juggler	1	2	3
A	D	E	F
B	E	F	D
C	F	D	E
D	A	C	B
E	B	A	C
F	C	B	A

The pattern version depends on whether circle rotations are aligned or opposite.

Prof Mike thought of this as two circles, but then we tried it (first on March 29, 2026 at MCS), and two triangles stand out. Mark Hayward recognized the `fast forward` symbol! Beginners on this pattern benefit from the placement of *advisory dots*!

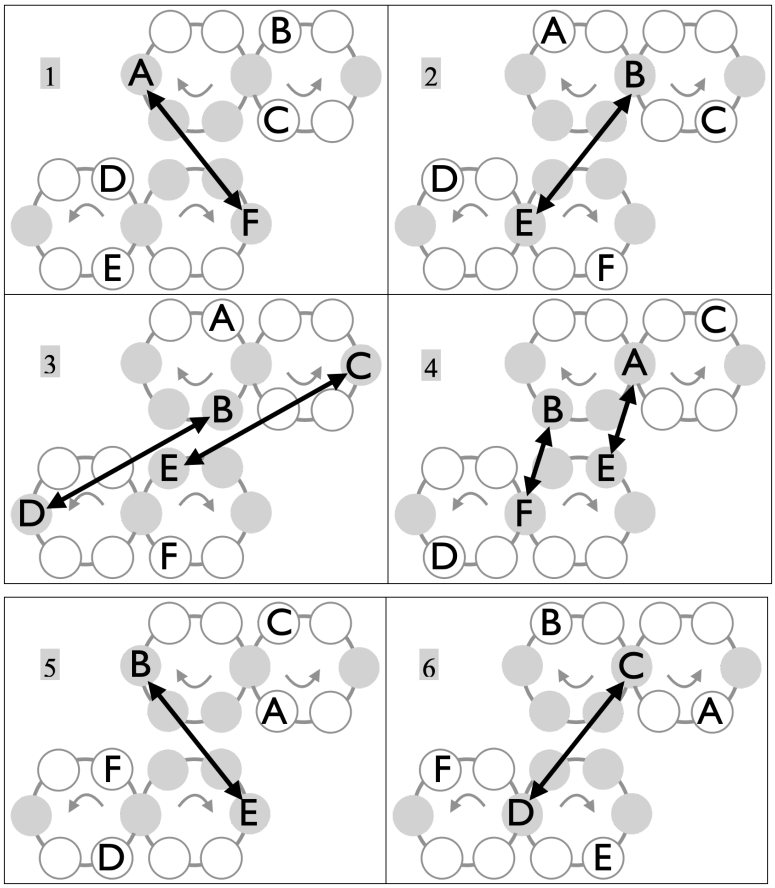
Metrics: J=6, CJ=3, pH=1, R=1, OJ=1 (v1) 3 (v2), PBJ=1/base, cycle=3 base count, motion =3.1 pu (if each circle has diameter 1 pu), and spin=0.

Foie Graham

This is an intense offset double-weave for six jugglers.
Pattern by Graham Paasch.

	Juggler					
Beat	**A**	**B**	**C**	**D**	**E**	**F**
1	**F**					**A**
2		**E**			**B**	
3		**D**	**E**	**B**	**C**	
4	**E**	**F**			**A**	**B**
5		**E**			**B**	
6			**D**	**C**		
7	**D**		**F**	**A**		**C**
8		**D**	**E**	**B**	**C**	
9			**D**	**C**		
10	**F**					**A**
11	**E**	**F**			**A**	**B**
12	**D**		**F**	**A**		**C**

Right-hand 2-count. In figure, pass in grey spots, move through open spots.

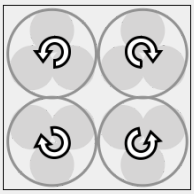


Jugglers **A**, **B**, and **C** move along one figure 8 while **D**, **E**, and **F** move along the other. As you enter an *outer* loop of your weave you do *pass self self* (right-hand counting) as usual for a weave. But as you enter the other loop, the *inner* loop, you make a series of 4 passes, starting and ending with the same person, and getting the other two in between. It takes advantage of the fact that coming out of the middle on a weave you are in front of your other weavers, and so can get some extra passes in before going to the backside of the weave loop. Thanks Conner for the pattern name!

Metrics: J=6, CJ=3, pH=1, R=1, OJ=3 , cycle=12 on 2-count, PBJ=¼. Motion=4 pu, like other weaves. Considering the swivelling back and forth, spin ≃ π rad per cycle.

Side-by-Side 2pi/3

Feels like a pattern we should have been doing for a long time!

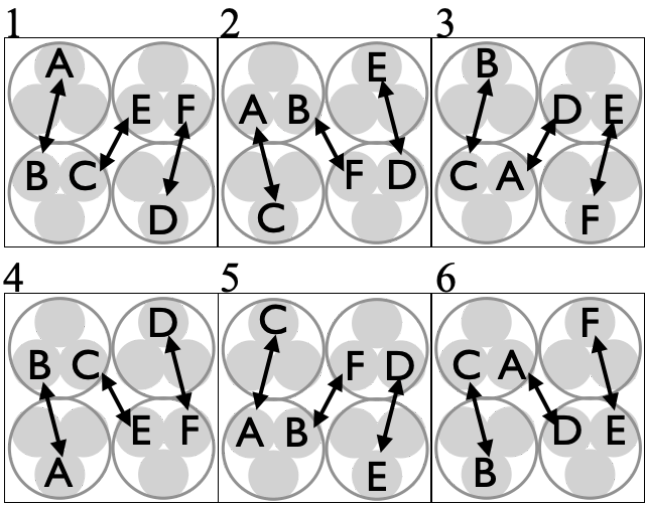


Constantly moving and passing on count of choice (e.g. 4, 5, or 6 are nice!)

Beat, 1 local cycle

Juggler	1	2	3
A	B	C	D
B	A	F	C
C	E	A	B
D	F	E	A
E	C	D	F
F	D	B	E

Pattern by Prof Mike



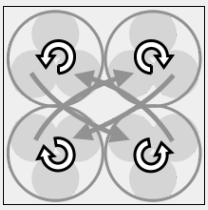
Refer back to the 3-person *2pi/3* or the base pattern, and notice that 2 such patterns can have the phantom passes synced up.

Move smoothly, staying in your circle.

Metrics: J=6, CJ=3, pH=1, R=1, OJ=3, PBJ=1/base, cycle=6 on base count, but passing sequence repeats after 3 beats. Derived from the base pattern, motion=4pu and spin=2π rad.

Interlocking 2pi/3

Big movement!

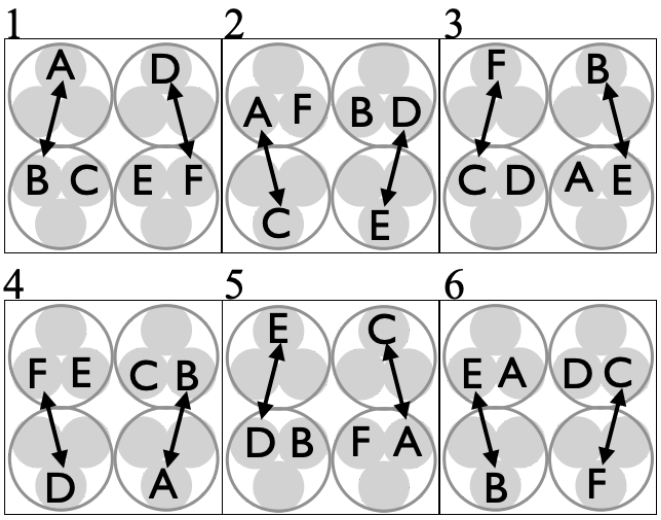


Compare set up to side-by-side…a lot more moving…a bit less passing

Beat, 1 local cycle

Juggler	1	2	3
A	B	C	
B	A		E
C		A	F
D	F	E	
E		D	B
F	D		C

Pattern by Prof Mike



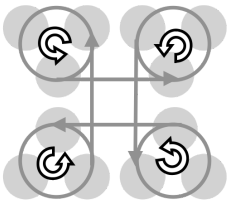
Refer back to the 3-person *2pi/3* or the base pattern, and notice that 2 such patterns can have the phantom passes synced up. **Rule**: switch sides on every such sync up.

Born April 26, 2026 at MCS.

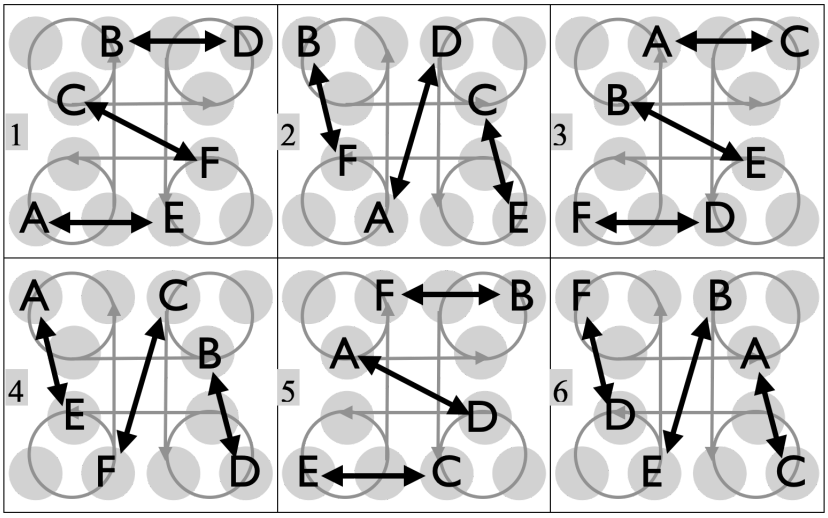
Metrics: J=6, CJ=3, pH=1, R=1, OJ=2, PBJ=2/(3*base), cycle=6 on base count, but passing sequence repeats after 3 beats. Spin inherits from the base pattern (spin = 2π rad), but motion is increased by the constant switching. Perhaps motion = 6pu per cycle.

Celtic Knot

A short cycle pattern with heavy-duty motion



	Beat, base count		
Juggler	**1**	**2**	**3**
A	**E**	**D**	**C**
B	**D**	**F**	**E**
C	**F**	**E**	**A**
D	**B**	**A**	**F**
E	**A**	**C**	**B**
F	**C**	**B**	**D**



Notice a knot-like trajectory (upper right) that connects 12 grey spots, 3 on each of 4 corners of the knot. The basic pattern has 6 jugglers position themselves at every other spot along that trajectory; they make a specific pass, and then move to the next spot. Various pass rules work, but the one shown above has a nice vibe. One way to think about it is to have three jugglers (**A**,**B**,**C**) set up in a line facing three others (**D**,**E**,**F**); passes are exchanged between opposite pairs. Then the middle two (e.g., **C** and **F** on beat 1 above) change sides by moving quickly through the middle. On completing their fly by, they each face to their right while rotating and initiating a backwards move into their respective corner loops, and then pass again (e.g. **C** with **E** and **F** with **B**), leaving the last pair (**D**, **A**) to be the new middle (beat 2 above), who pass and change places as the pattern progresses. Remember the British driving rule when you move through the middle: keep your passing partner on your right as you walk past!

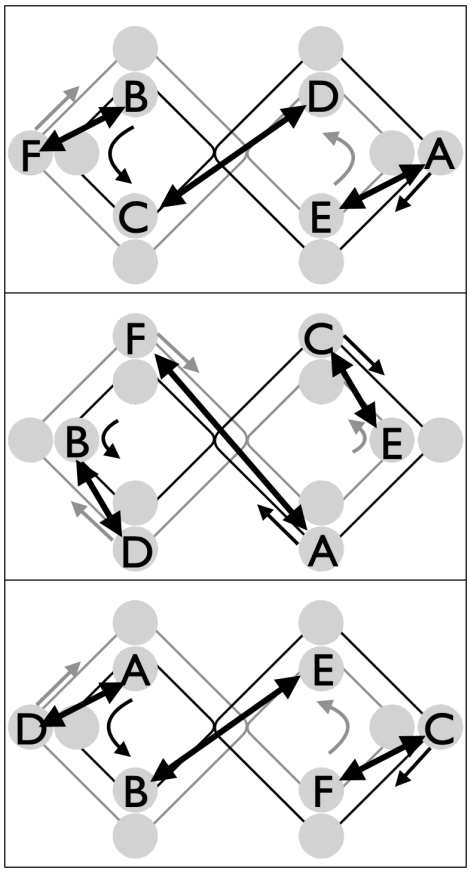
Metrics: J=6, CJ=3, pH=1, R=1, OJ=3, full cycle=12 on base count; local passing cycle is 3 beats. 6-count base is a good start, but 5-count is fun, and 4-count takes some smooth moves! PBJ = (⅓)(1/base). Assuming about 2pu per corner and 1pu in between corners, we get motion = 12pu per cycle. Likewise, there's about a ¼ turn rotation on every beat, giving an overall pattern spin=6π rad per cycle. Pattern by Prof. Mike; first attempted at the 2024 IJA festival in Green Bay. It takes some practice to control your own pattern well enough to move quickly through the middle without colliding with your counterpart and while prepping for the rotations needed to complete the cycle. Performed at 2025 Madfest show: Goose was there!

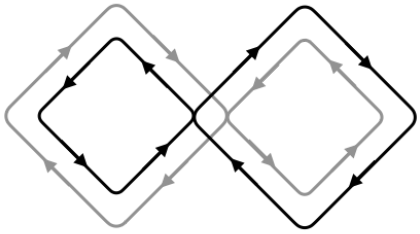
Slippery Knot

Take the knot up a notch!

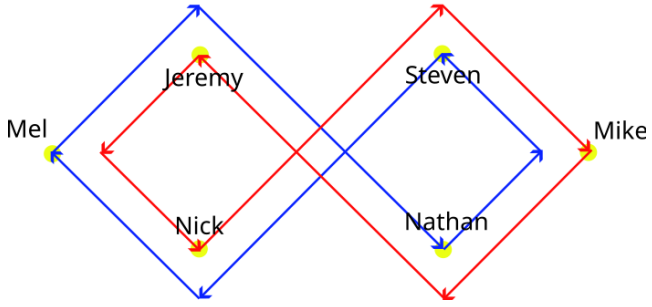
	Beat, base count		
Juggler	**1**	**2**	**3**
A	**E**	**F**	**D**
B	**F**	**D**	**E**
C	**D**	**E**	**F**
D	**C**	**B**	**A**
E	**A**	**C**	**B**
F	**B**	**A**	**C**

The *Slippery Knot* is not for beginners. Be fleet of foot, juggle slow and move fast! It looks pretty cool! By Prof. Mike and Nathan Nard.





From the *Celtic Knot,* we asked what happens if all three pairs would switch sides after passing, instead of just the middle pair doing so. We discovered this heavy-duty moving pattern. All jugglers move around two squares, with (**A**,**B**,**C**) following one path (black), and (**D**,**E**,**F**) the other, opposing path (grey). The rule is pass then move to the next spot. Perhaps the trickiest part is making the long pass and long move through the middle, all while avoiding the nearby jugglers. Notice too that there's one small square and one bigger square, so some corners are sharper than others. Also, British driving rules apply!

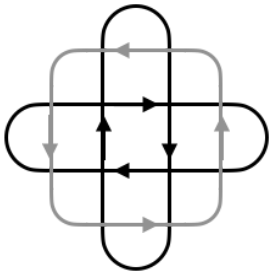


Nathan Nard's set up from the 2026 Madfest show. The all-pass all-move effect is impressive when it works!

Metrics: J=6, CJ=3, pH=1, R=1, OJ=3, full cycle=6 on base count; local passing cycle is 3 beats. 6-count base is a good start, but 5-count is fun, and 4-count takes some smooth moves! PBJ = (⅓)(1/base). Motion=8pu, assuming average edge is 1pu. Likewise, about 6 ¼ turns are needed, making, spin≃ 3π rad per cycle. Performed by Nathan, Mike, Nick, Steve, Mel, and Jeremy in the 2026 Madfest show.

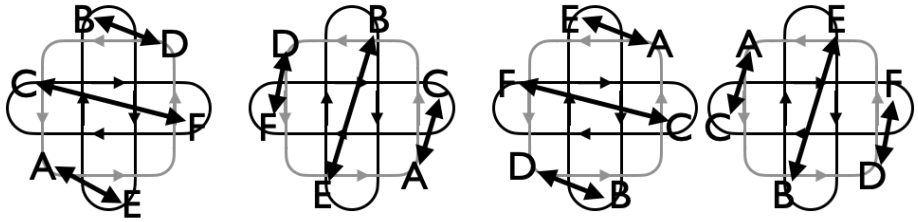
Disco Knot

Another constant motion pattern



	Beat, base count	
Juggler	**1**	**2**
A	**E**	**C**
B	**D**	**E**
C	**F**	**A**
D	**B**	**F**
E	**A**	**B**
F	**C**	**D**

The pass-move rule has everyone flying by everyone else, but only needing to track two passing partners each.



Though the figure doesn't suggest it, the pattern comes from an always-switch-sides version of the *Celtic Knot,* while maintaining the turn rule. It helps to set it up with *advisory dots*!

Metrics: J=6, CJ=3, pH=1, R=2, OJ=2, full cycle=4 on base count; local passing cycle is 2 beats. 6-count base is a good start, but 7 and 5-count are fun. PBJ = (1/base). Assuming sides average to 1pu, pattern motion = 4pu. Four quarter turns makes the pattern spin=2π rad. Conjured by Prof Mike on May 4, 2026, and first juggled a few days later on Library Mall by Mike, Mark, Nathan, Chan, and Nick. Mark noticed the shape's relationship with disco rectangles! Movement on the two disco rectangles is in the opposite direction to movement on the square, creating a lot of chaos! Passes alternate north-south then east-west, which minimizes club-collision risk, though body-collision risk is high!

Meltic Knot + Orbiter

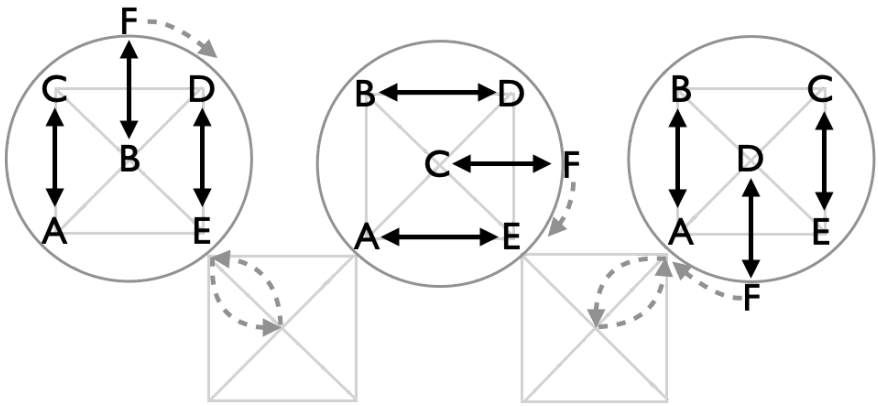
Best with a ripsticker!



Beat, base count (6-count is standard)

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A	**C**	**E**	**B**	**D**	**F**															
B	**F**	**D**	**A**	**C**	**E**															
C	**A**	**F**	**E**	**B**	**D**															
D	**E**	**B**	**F**	**A**	**C**															
E	**D**	**A**	**C**	**F**	**B**															
F	**B**	**C**	**D**	**E**	**A**															

Local cycle = 5 beats
Full cycle=20 beats

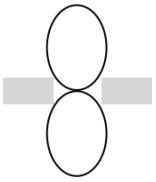


Three beats shown. **B** and **C** swap after 1, then **C** and **D**, etc.

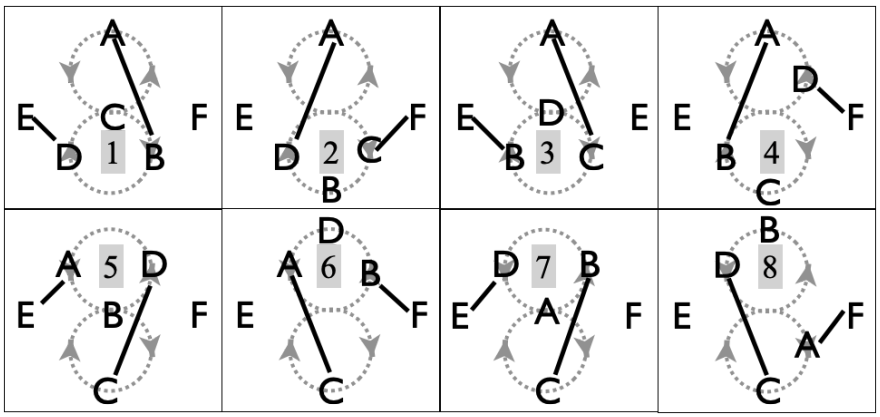
Refer back to the 5-person *Meltic Knot* for base pattern, which is extended here by adding one orbiter (**F**) who picks up all the phantom beats. Because those beats show up on the base count and give openings exterior to the 4 sides of the pattern, it takes a very fast juggler to do the orbiter role; we solve that by putting a ripsticker in as **F**. They lead the up-down to get us off to a proper start. Even forgetting the move in and out of the middle (shaded box and arrow), you are constantly swinging back and forth through a ¼ circle to pass to your neighboring corner. Making good leads for the ripsticker is important

Metrics: J=6, cycle=20, CJ=3, pH=1, R=2, OJ=5, PBJ=⅙, on 6-count, but generally 1/base. With sides about $\sqrt{2}$ pu on the box, the 5 central jugglers each covers about 8 pu to get back to the start. With **F** on a circle of radius $\sqrt{2}$ pu, the ripsticker completes 5 full turns per cycle, for distance $10\sqrt{2}\pi$ pu, and thus for a pattern average of about 14 pu per cycle! There's also a lot of ¼ turns to alternate corners for passing, making total spin = 6π rad for internal jugglers, and 10π for **F**, making pattern average spin 3 ⅓ full turns (1200°; 20.9 rad) per cycle. Pattern by Mellonhead and Prof Mike. See video from Luke's loft with Jeremy as the ripsticker!

El Turbo with 2 posts



	Beat, base count: 2-count rights is standard; 3-count is pleasant!															
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
A	B	D	C	B	E	C		F	B		E	B		F	D	
B	A		E	A		F	C		A	C	D	A	E	D		F
C		F	A		D	A	B	D	E	B		F	D		E	D
D	E	A		F	C		E	C		F	B		C	B	A	C
E	D		B		A		D		C		A		B		C	
F		C		D		B		A		D		C		A		B

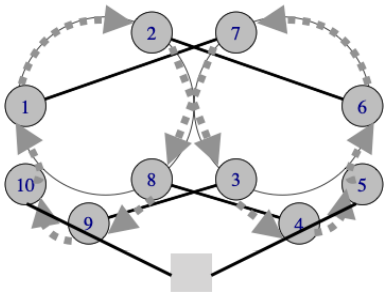


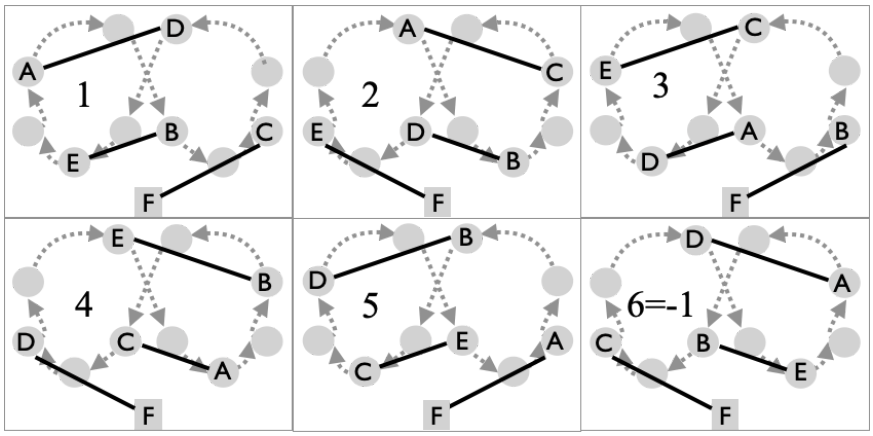
Metrics: J=6, cycle=16, PBJ=(⅔)(1/base), CJ=3, OJ=4 ⅔ , pH=1, R=2. Movers each travel 5pu, for a pattern motion (⅔)5pu. Movers also rotate a full turn and the posts swivel a ½ turn per cycle, for a pattern spin of 1.7π rad.

½ cycle shown. **A, B, C,** and **D** do the *El Turbo,* and **E** and **F** are the posts picking up some selves. But here there's either an odd-beat post (**E),** or an even-beat post **(F).** Pattern by MAJ. See *El Turbo* (4-person) for more on the base pattern. As in *El Turbo*, each juggler feeds 4 spots (with #1 and #4 the same person), before moving on. Those feeds are highlighted (grey shading) in the passing table.

Ivy Pi Weave + Post

There are so many ways to move and juggle through a figure 8!





See 5-person *Ivy Pi Weave* for the base pattern.

	Beat, base count (4-count standard)				
	1	**2**	**3**	**4**	**5**
A	**D**	**C**	**D**	**C**	**F**
B	**E**	**D**	**F**	**E**	**D**
C	**F**	**A**	**E**	**A**	**E**
D	**A**	**B**	**A**	**F**	**B**
E	**B**	**F**	**C**	**B**	**C**
F	**C**	**E**	**B**	**D**	**A**

Metrics: Inherits mainly from *Ivy Pi Weave*. J=6, full cycle 10 beats, local cycle 5; PBJ=¼ (on 4-count base), CJ=3, OJ=3, R=2. The base pattern has high spin, 5π rad per mover, but a bit more rotation is required here to accommodate passes to **F**; likewise **F** swivels back and forth 5 times per cycle. Splitting the difference, the pattern average spin is close to 5π rad. Motion about 6pu per cycle per mover, for a pattern motion = 5pu.

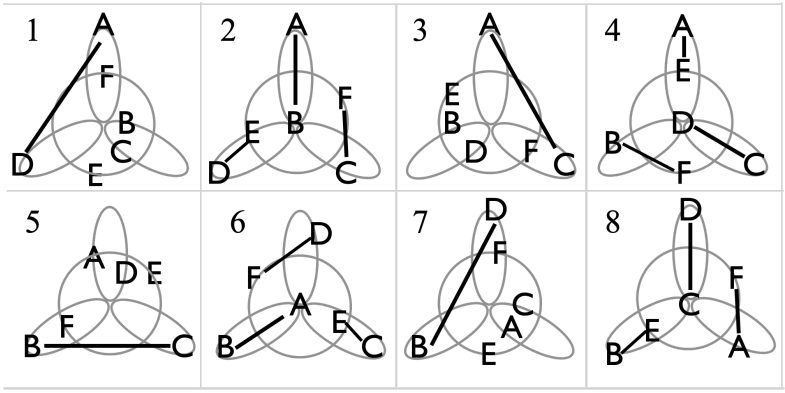
We usually juggle on a 4-count base to keep the moves doable. So, e.g., each juggler passes 4 in a row on that count, and then once to **F**. But each mover only passes to 2 other movers, while everyone passes to the post. Note the **sharp turns** going from positions 2-to-3 and 7-to-8.

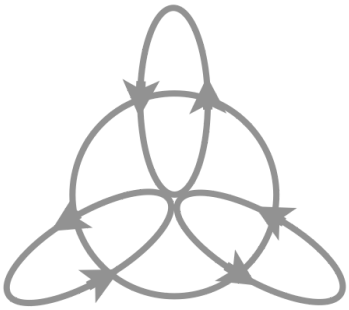
Pollinator

aka, *Three Leaf Clover with 2 Orbiters*



Revisit the 4-person *3-leaf clover*, but add two jugglers (**E** and **F**) to absorb those pesky self throws! In this version, the orbiters aren't really 'orbiting' at all, they are smack in the middle of the pattern. As drawn, **E** and **F** move clockwise (from above) along the perimeter of an inner circle. Orbiter **E** passes to the person who has just finished their feed (initially **D**), then on ½ the count of the others, **E** passes to **A, C, B,** repeat. Orbiter **F** passes at the same pace as **E,** and passes to the person who is about to start their feed. Then the clover-movers do the standard 3-leaf clover, but there are two more jugglers packed into the space, increasing the pattern's molecular weight! There are also fewer self throws per cycle, as the orbiters pick up 2 passes per other-mover per local cycle, making the rule a *1-in-1-out clover.* 2-count is doable but 3-count is much more relaxing!

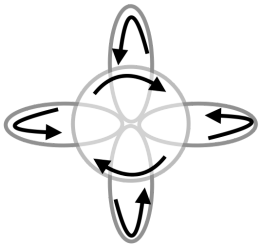




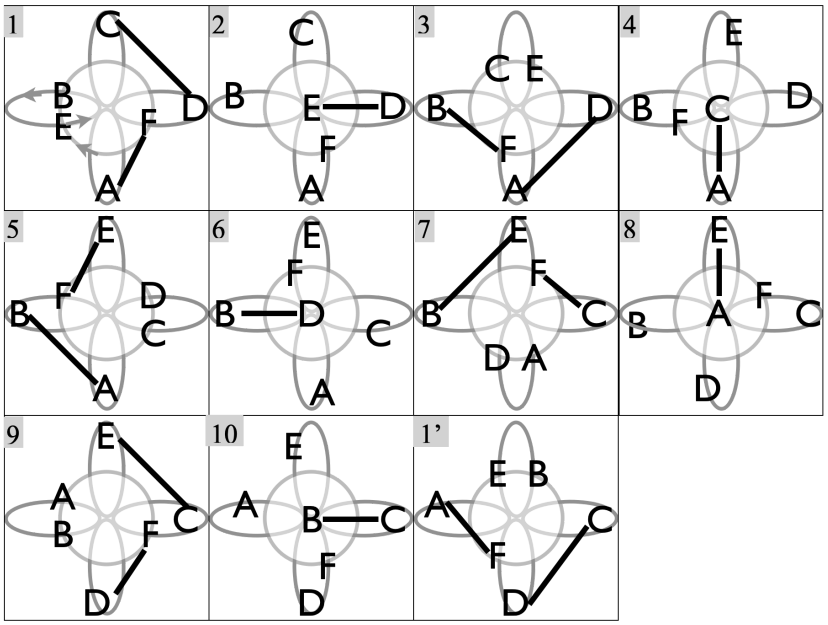
	Beat, 3 count base is ideal. Feeder is grey.							
	1	**2**	**3**	**4**	**5**	**6**	**7**	**8**
A	**D**	**B**	**C**	**E**		**B**		**F**
B		**A**		**F**	**C**	**A**	**D**	**E**
C		**F**	**A**	**D**	**B**	**E**		**D**
D	**A**	**E**		**C**		**F**	**B**	**C**
E		**D**		**A**		**C**		**B**
F		**C**		**B**		**D**		**A**

Metrics: J=6, local cycle=8; full cycle=24 to get back to original spots;, base count = 3, OJ=5, CJ=3, pH=1, R=2, PBJ=2/9 (3-count base). Motion inherits from *Three Leaf Clover* (5 pu and 2.5π rad per full cycle), then including the two orbiters gives pattern average motion at about 7pu. Likewise the average spin is about 4.3π rad per cycle, considering the circle traversed an equivalent of 4 times by orbiters **E** and **F** over the full cycle.

Four leaf clover with 1 orbiter



Add one juggler to the 5-person *4-leaf clover* to absorb some of those pesky self throws! In this version, orbiter **F** moves clockwise (from above) along the perimeter of an inner circle.



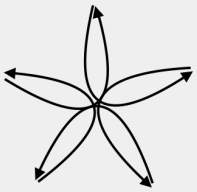
Recall the 10 beats of a *4-leaf clover* entail a feed of 3 positions, then 2 phantom beats to move into the middle, one pass, and 4 phantoms out to the leaf on your left. In this variation, orbiter **F** passes to the person who has just gone through the middle, and is on the 3rd self of their 4 phantom out beats. The clover-movers do the standard 4-leaf clover, but with one more juggler packed into the same space! In the table, grey indicates feeder.

	Beat (base count; 2 or 3- count)									
	1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
A	**F**		**D**	**C**	**B**			**E**		
B			**F**		**A**	**D**	**E**			**C**
C	**D**			**A**			**F**		**E**	**B**
D	**C**	**E**	**A**			**B**			**F**	
E		**D**			**F**		**B**	**A**	**C**	
F	**A**		**B**		**E**		**C**		**D**	

Metrics: J=6, local cycle=10, full cycle=40 to get back to original spots. Base count=2 or 3, OJ=5, CJ=3, pH=1, R=2, PBJ=¼ (2-count), ⅛ (3-count). For the clover, full cycle motion is 7pu and spin is 3π rad per cycle. The orbiter, meanwhile completes about 3pu per loop, and 4 loops per cycle, for 12pu, giving a pattern average motion near 8pu and a pattern average spin close to 8π rad per full cycle.

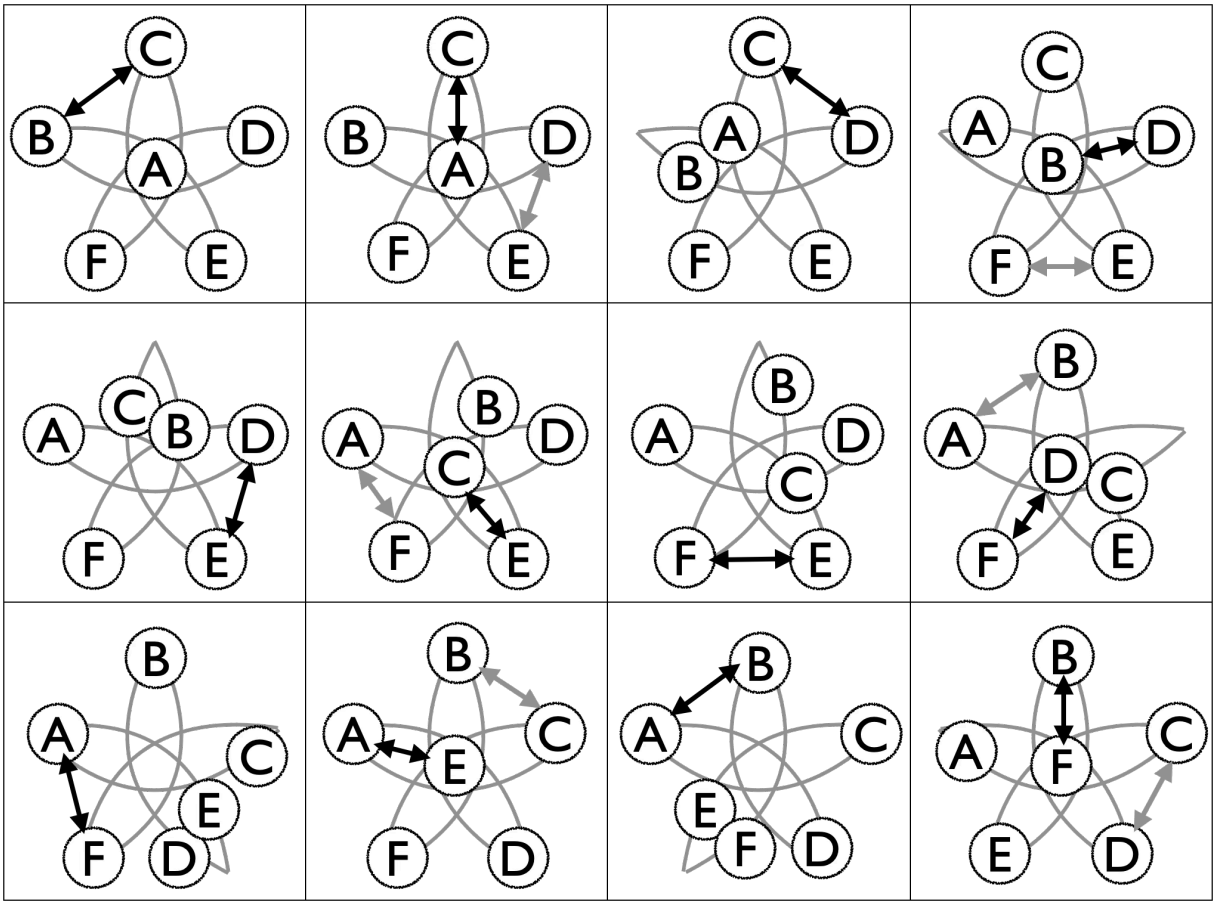
Variations arise by changing the orbit rule, such as by passing on the 2nd rather than 3rd beat out, or by picking up the tail or the head of each feed segment. The version shown seems to have more staying power!

Five leaf clover



Among the multitude of options, this 2-in-6-out is the natural extension of lower-leaf clovers and one we like, especially when we add the extra pre-feed pass (grey in figure)!

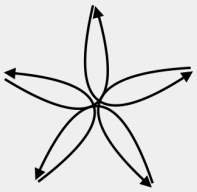
in



	Beat, base count (2 or 3 count); feeder position in grey											
	1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**
A		C				F		B	F	E	B	
B	C			D				A		C	A	F
C	B	A	D			E				B		D
D		E	A	B	E			F				C
E		D		F	D	C	F			C		
F				E		A	E	D	C			B

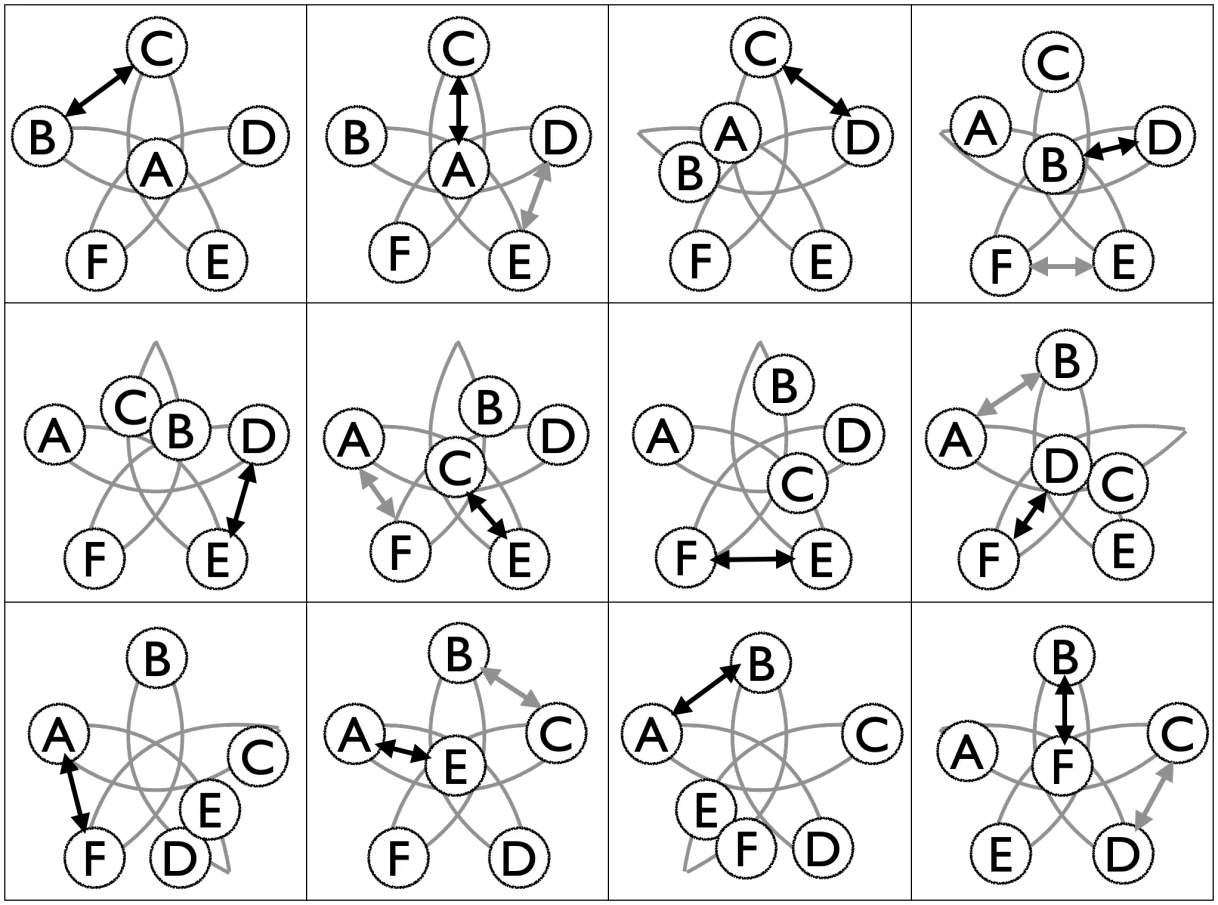
Metrics: J=6, CJ=3, pH=1, R=1, OJ=4, PBJ=⅛ (2-count base) or 1/9 (3-count base). With the pre-feed pass, it's ¼ and ⅛, respectively. We say the cycle is 12 beats, though 5 such cycles are needed to get back to the starting positions. Motion/spin ..high

Five leaf clover



Among the multitude of options, this 2-in-6-out is the natural extension of lower-leaf clovers and one we like, especially when we add the extra pre-feed pass (grey in figure)!

in



Comment on 2 in 6 out and also to 2 in (3-p-1-p)-out ; compare with Reddit posted video from July 2026…which is a non-cycleable 4-in-4-out variation, it seems!!

Double Vision 2pi/3

Is this a pattern?

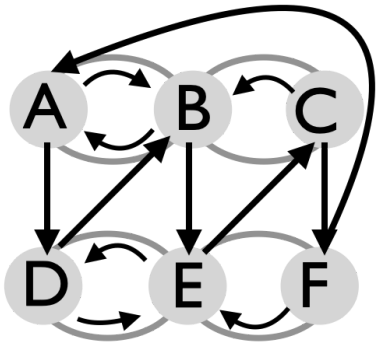
Beat, 1 local cycle			
Juggler	**1**	**2**	**3**
A			
B			
C			
D			
E			
F			

Pattern by Prof Mike

Metrics: inherits from 2pi/3…but make more motion by root or so; go on 3 4 or 5 cnt

Extra Club Moving Six Person Patterns

Extra Club Double Weave



1-count base		Juggler					
		A	B	C	D	E	F
	1	**D**					
	2				**B**		
	3		**E**				
	4					**C**	
	5			**F**			
	6						**A**

(**A**,**B**,**C**) start right-handed; (**D**,**E**,**F**) start out of phase, each with 2 clubs in the left hand. All passes are doubles. Set up a standard double weave, but have one juggler (**A**) with an extra club. Passing beat stays at 6-count; moving stays the same (figure eight), but the passes are all doubles. Importantly, you receive from one juggler then pass to another, in order to retain the 7-club-style passing. Inspection shows that the passing sequence is the same (relabeled) as the stationary *Trompe l'oeil.* Leads are really critical in this pattern; lead deeply on the ends and short and lofty in the middle. The passing rate and moving rate are well within the skills of good movers/jugglers, but still the pattern eludes some perfectly capable groups.

Metrics: J=6, CJ=3 ⅛, pH=2, R=1, local passing cycle=6 on 1-count; full cycle 24 on 1-count base; PBJ=⅛, motion=4 (same as double weave), spin is low.

Variations: Start in *Trompe l'oeil* formation, then do 2 passes followed by a move into a circle; followed either by a move back or by an *Oogle-boogie* style move sequence. Tried on June 7, 2026 by Luke, Mike, Nathan, Chan, Gabe, and Keenan. Less common, but doable, is to weave via the dresser drawer style!

Double Gandini Weave Plus

There is so much time at the back end of the Double Gandini Weave! Nine self throws! There must be something to do in all that time. Wait, I know! Let's add another club!This is the same as the previous pattern with a triple club passed back and forth over the top of the pattern. It's a spectacular pattern and not as hard as you might think (though certainly not easy).

Juggler/beat	**1**	**2**	**3**	**4**	**5**	**6**
A		T-F		E	F	
B		D	E			T-E
C	D			T-D		F
D	C	B			T-B	

| **E** | T-A | | B | A | | |
| **F** | | | T-C | | A | C |

> * Right-hand countingPattern by Luke Emery.

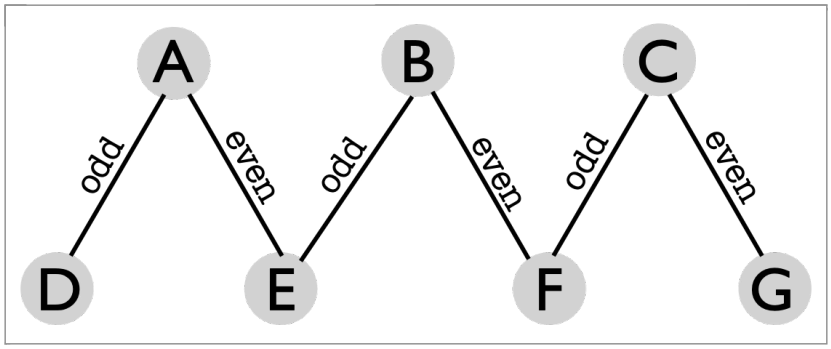
Seven Person Patterns

Stationary Seven Person Patterns

3 v 4 lines

Maybe the easiest entry-level passing pattern for 7.

Beat, 2 or 4 count base		
	1	2
A	D	E
B	E	F
C	F	G
D	A	
E	B	A
F	C	B
G		C



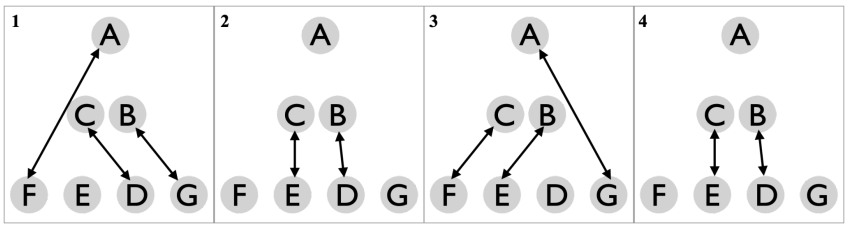
We juggle this in Madison only when a bunch of newby club passers want to be in a biggish pattern. It's fun and good practice for timing; and it scales easily with more or fewer jugglers.
Magic club variation: Have the first club released by **D** to be of a different color than the other 20 clubs. Try to keep the pattern going as **the magic club** moves all the way from one side to the other. An engaging game!

Metrics: J=7, CJ=3, pH=1, OJ=12/7, R=2, cycle=2 beats on base count (usually 4-count for beginners), PBJ=3/14 (on 4-count base), motion=0. Noting that 5 of the 7 passers are swivelling as in an ordinary feed of two, we get per feeder spin of $\pi/3$ rad, and a pattern average $(\pi/3)(5/7) \approx 0.75\,rad$, which is about 43° (which is near zero per beat).

1-2-4 Apollo

A nice and easy MAJ entry-level passing pattern.

	Beat, 2-count base			
	1	2	3	4
A	F		G	
B	G	D	E	D
C	D	E	F	E
D	C	B		B
E		C	B	C
F	A		C	
G	B		A	



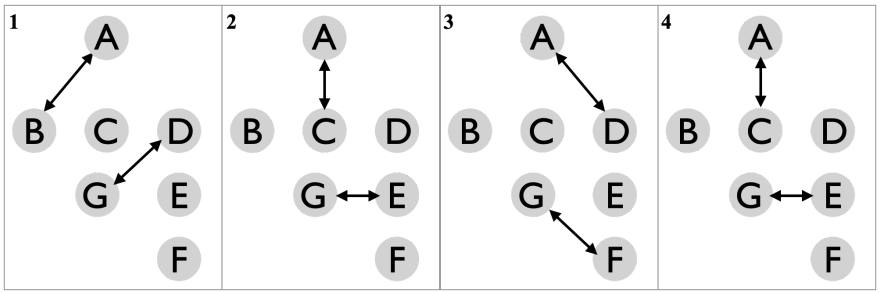
Corners **A**, **F, G**, each pass on 4-count, alternating to two spots. Sweepers **B** and **C** each do standard sweep feed of three positions. Finally, **E** and **D** have interesting sweep-feed-like positions, each with a gap. We call that gap a *phantom*; it's convenient to explain that someone is doing a sweep feed of 3 positions, with one of their feedees missing. To keep the pattern interesting, repeat tries to get a good run, then move everyone around and repeat. Try to get through all 7 positions!

Metrics: J=7, CJ=3, pH=1, OJ=16/7, R=4 (recognizing that **A** is different from **F** and **G**), PBJ=5/14, cycle=4 beats on 2-count base, motion=0. Referring to spin levels for basic feeds, **A**, **B**, and **C** spin about $2\pi/3$ rad and the others about $\pi/3$ rad, for a pattern average near $\pi/2$ rad (i.e, a ¼ turn). **ASK MARK ABOUT THE ORIGIN**

Hinge

Two simply connected feed lines make a nice and easy entry-level passing pattern for 7 jugglers.

	Beat, 2-count base			
	1	2	3	4
A	B	C	D	C
B	A			
C		A		A
D	G		A	
E		G		G
F	G			
G	D	E	F	D



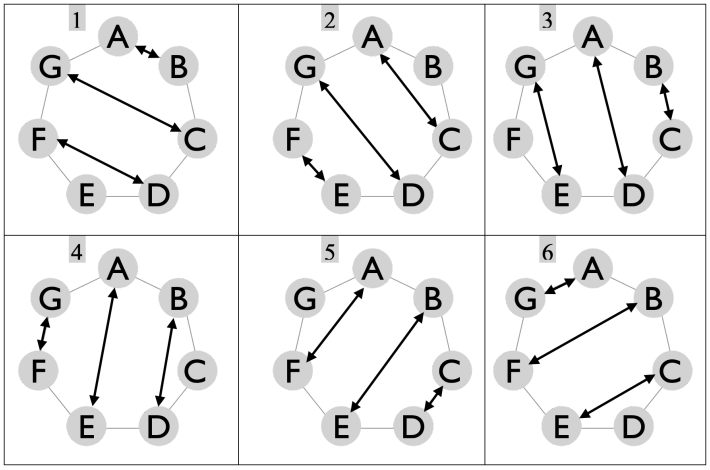
Feeders **A** and **G** perform sweep feeds of respective lines of three jugglers (**B**,**C**,**D**) and (**D**,**E**,**F**). We call the overlapping position, **D**, the *hinge,* as they swivel back and forth on a 4-count between the two feeders.

Metrics: J=7, CJ=3, pH=1, OJ=12/7, R=4, PBJ=2/7, cycle=4 beats on 2-count base, motion=0. Feeders A and G spin about $2\pi/3$ rad per cycle, about the same as the hinge D, giving a pattern average spin=$2\pi/7$ rad, even less than the *Apollo.* ?origin?

Feast for 7

A popular stationary pattern for 7 jugglers.

Everyone feeds everyone else in this classic pattern: go around the circle on any prespecified count, and be sure to include yourself in the count.



Base count	Juggler						
	A	**B**	**C**	**D**	**E**	**F**	**G**
1	**B**	**A**	**G**	**F**		**D**	**C**
2	**C**		**A**	**G**	**F**	**E**	**D**
3	**D**	**C**	**B**	**A**	**G**		**E**
4	**E**	**D**		**B**	**A**	**G**	**F**
5	**F**	**E**	**D**	**C**	**B**	**A**	
6	**G**	**F**	**E**		**C**	**B**	**A**
7		**G**	**F**	**E**	**D**	**C**	**B**

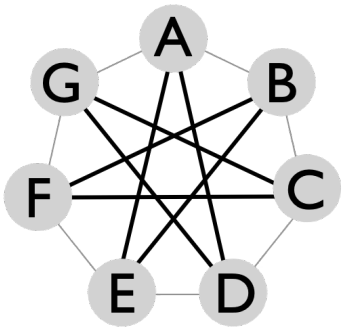
Full cycle is 7 beats on the base count (6 beats are shown in figure). Self throws shaded in table (right). While the pattern sets up as a stationary pattern, larger feasts like this *Feast for 7* work well when jugglers slowly oscillate so as to shorten long passes and extend shorter passes. This leads to a pleasant *amoeba* effect! On counts, we often do 4 or 3-count for a friendly cycle, though 2-count rights is really more fun. When you're comfortable with the 3-count version, try the *Picnic* variation in which each juggler completes a cycle of all crossing passes (e.g. right-right or left-to-left) followed by a cycle of all straights; the rule is to switch pass type as you move through your self beat. In a *Picnic* start, **A**, **G**, and **F** would start crossing and **B**, **C**, and **D** would start straight.

Metrics: J=7, CJ=3, OJ=6, pH=1, R=1, cycle=7 on any base count, PBJ=(6/7)(1/base) (e.g. 2/7 on 3-count base). motion=0. Considering the interior angles of a regular heptagon, jugglers swivel $10\pi/7$ rad per cycle (about 260°) (and more if you pirouette on your self beat!).

7-point star

It's about timing!

	Beat	
	odd	even
A	E	D
B	F	E
C	G	F
D	A	G
E	B	A
F	C	B
G	D	C



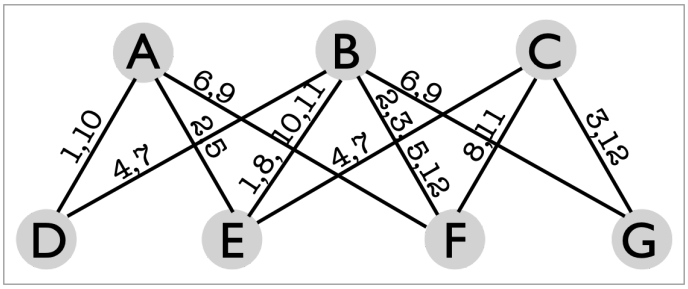
This one is not for beginners, in spite of having no motion and essentially no spin! Jugglers set up in a circle, and depending on the chosen base count, they pass three spots to the right (on right throws) and three spots to the left on left throws. Collisions are avoided when everyone is well synced up: e.g. on a right-hand release, **A**'s pass to **E** goes just behind **G**'s pass to **D** and ahead of **B**'s pass to **F** and **C**'s pass to **G**! But it takes precision to stay synced!

Metrics: J=7, CJ=3, pH=1, R=1, OJ=1 (even count) 2 (odd count), PBJ=1/base, cycle length=1 (even count) 2 (odd count). Best to try 2, 3, or 4 base count. Motion=0, spin=0 on the even count and a very marginal $2\pi/7$ rad, or about 51°, noting the swiveling back and forth on odd base counts.

Overlapping Chocolate Bar 3-count feeds

A great stationary pattern for mixed skill levels, including one super challenging spot and lots of passing from both hands.

Recall the 5-person *Chocolate Bar 3-count Feed,* which has two chocolate-bar feeders and three 3-count feedees. This extension for seven overlaps two copies of the classic by merging two offset chocolate bars into one central juggler (**B**), and merging adjoining 3-counters into double-3-count spots (**E**,**F**). The passing itself requires typical MAJ precision, but the real challenge is remembering who you're passing with! In fact, juggler **B** does a pretty special 1-count. **A** and **C** continue with chocolate bar; **D** and **G** do 3-count; and **E** and **F** do 3/2-count.



Passing beats are indicated on the edges

Base count	Juggler						
	A	**B**	**C**	**D**	**E**	**F**	**G**
1	**D**	**E**		**A**	**B**		
2	**E**	**F**			**A**	**B**	
3		**F**	**G**			**B**	**C**
4		**D**	**E**	**B**	**C**		
5	**E**	**F**			**A**	**B**	
6	**F**	**G**				**A**	**B**
7		**D**	**E**	**B**	**C**		
8		**E**	**F**		**B**	**C**	
9	**F**	**G**				**A**	**B**
10	**D**	**E**		**A**	**B**		
11		**E**	**F**		**B**	**C**	
12		**F**	**G**			**B**	**C**

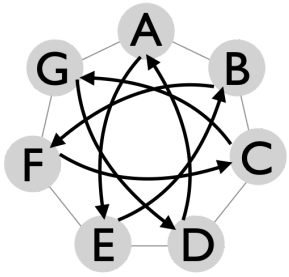
The passing sequence is pretty elaborate, but it follows nicely after you practice the separate 5-person sub-patterns. I.e., to start, ask **C** and **G** to sit out, and with the remaining 5 practice the *chocolate-bar 3-count feed* (passes indicated in gray in the table above). When you're clear on the passes in that 12-beat cycle, then ask **C** and **G** to rejoin, and at the same time ask **A** and **D** to sit out; this creates a 2nd *chocolate-bar 3-count feed* in which, for example, **B** starts pass-pass, respectively to **E** and **F** (passes indicated by no shading in the table). With both sub-patterns ready, it's time to put the patterns together.

Metrics: J=7, CJ=3, OJ=20/7, pH=1, R=4, cycle=12 on 1-count base, PBJ=4/7, motion=0. The spin calculation captures all the swivelling needed to cover the 12 beat cycle. From standard feed coordinates, we get **B**'s spin itself is over a full circle, and the pattern average spin equals 2.5 rad (about 145°). **B**'s spot must be one of the most difficult non-moving non-extra-club spots in the repertoire.

Extra Club Stationary Seven Person Patterns

7-count 7-point Star

Serenity with 7.



Jugglers are in sync and in phase, with 1 extra club that triggers double crossing throws, one on each count from 1 to 7.

Surely not an MAJ origin, but a really good pattern!

1-count base		Juggler						
		A	B	C	D	E	F	G
	1	**E**						
	2					**B**		
	3		**F**					
	4						**C**	
	5			**G**				
	6							**D**
	7				**A**			

Metrics: J=7, CJ=3 1/7, OJ=1, pH=2, R=1, PBJ=1/7, cycle=7 on 1-count, motion=0, spin=0.

Double N+, with 1 cap

The 7-person extension of the Graham/Claude N+ with 2-count double-singles passing

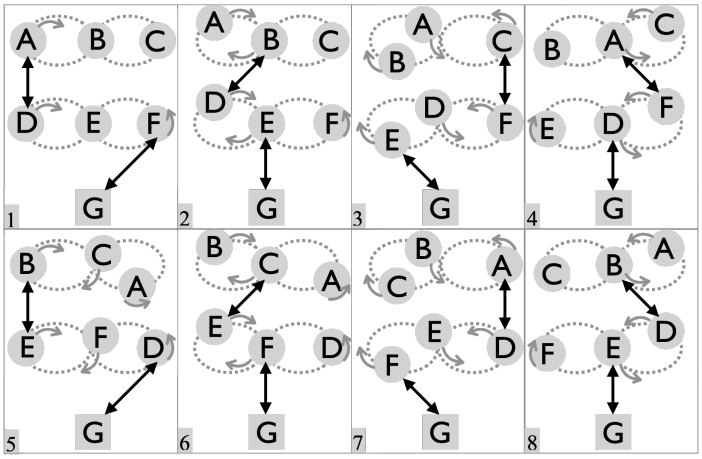
1-count base		Juggler						
		A	B	C	D	E	F	G
	1	**E**						
	2					**B**		
	3		**F**					
	4						**C**	
	5			**G**				
	6							**D**
	7				**A**			

Metrics: J=7,

Moving Seven Person Patterns

Open face sandwich with olive

3-3-1 Twister



To appreciate the name, it's good to look ahead to the 9-person *Twister Sandwich,* which organizes jugglers into three stacked weaves, the outer two like pieces of bread and the inner one the filling, or in code, a 3-3-3. This 3-3-1 replaces one slice of bread with a single stationary juggler, the *olive (**G**),* giving an open-face sandwich metaphor. Jugglers **(A,B,C)** perform a standard weave, each passing to two other twisters, among **(D,E,F)**, who perform a double-time twister, moving swiftly through a figure 8 with a single right self on each direction change.

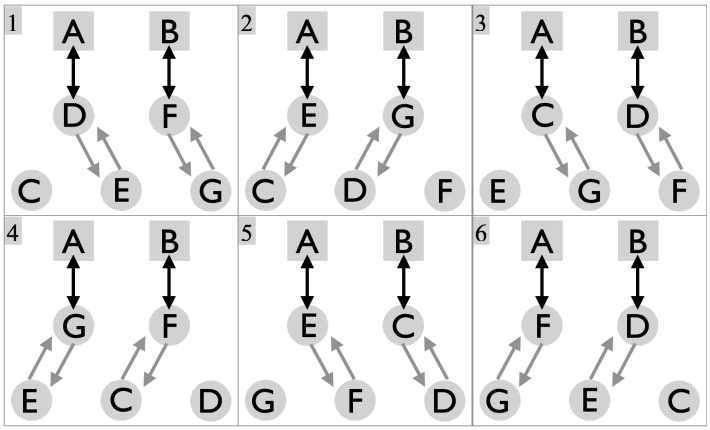
Base count		Juggler						
		A	B	C	D	E	F	G
	1	D			A		G	F
	2		D		B	G		E
	3			F		G	C	E
	4	F			G		A	D
	5		E		G	B		D
	6			E		C	G	F
	7	D			A		G	F
	8		D		B	G		E
	9			F		G	C	E
	10	F			G		A	D
	11		E		G	B		D
	12			E		C	G	F

Variation: The most difficult positions above are the double-time twisters **D**, **E**, **F**, as they have just one right self to turn after passing twice to each side. A nice variation lessens their work by having them do an ordinary (single-time) twister: *pass-self-self-pass-self-self-etc.* Then **G** drops to a standard 4-count Doctor, and the only issue is what to do about **A, B, C**. Solution! Get **A, B, C** doing a 4-count 2 pi/3, which you recall has a 4-count opening to plug nicely in. In effect each person in that slice of bread passes to the other two in their slice and also to one juggler in the middle of the sandwich, all on 4-count. This variation mixes two nice but different ways of having 3 jugglers move through the figure 8: twister and 2pi/3. Perhaps in shorthand, the original is: *weave;double-time-twister;2-count Doctor*, and the variation is *2pi/3; twister; 4-count Doctor.* Naturally, other weave shapes can replace the **A-B-C** weave in the original; for instance an inverted weave or a dresser drawer weave, but these are less popular. Yet another variation from the original just drops **G** to 4-count and asks twisters **D, E, F** to be lopsided, passing twice to the **A-B-C** weave line and just once to **G** on each turn: *pass-pass-self-pass-self-self,* say, on the base count.

Metrics: J=7, CJ=3, pH=1, cycle=12 on 2 count, R=3, OJ=18/7=2.6, PBJ=2/7=0.29, motion = 6.9 pu, counting each mover as covering 4 pu per cycle. From the components we get an average spin just shy of 2π rad (1 full circle). (Remember, the twisters are maxed at about 4π rad per cycle.) The *2pi/3; twister; 4-count Doctor* variation has similar metrics; OJ changes per juggler but averages the same at 2.6. , PBJ=3/14=0.21 (slightly easier), and spin increases to about 2.7π rad.

The Blast

Perfect for 5 good movers and 2 4-count feeders.



Feeders **A** and **B** set up side by side facing all the others, who are organized in two rows. 4-count is the standard base count, and all passes happen between the feeders and the two jugglers occupying the front row. After passing, the front two movers (e.g. **D** and **F** to start) back up, while two of the back movers move forward (**E** and **G** to start); the other mover sticks on this beat. Between each passing beat 4 jugglers move (swiftly!), all swapping spots, and one sticks. Poor spacing and juggler collisions are common with beginners. Mel likes to recommend the front movers push straight back after the pass and then shift, while Mike favors recommending an angle! Be sure to push forward especially briskly when coming out of the middle back position. In order to keep track of the passing rhythm, MIke also likes to think of movers as doing 4-count with a bunch of phantoms in the back line.
performed when

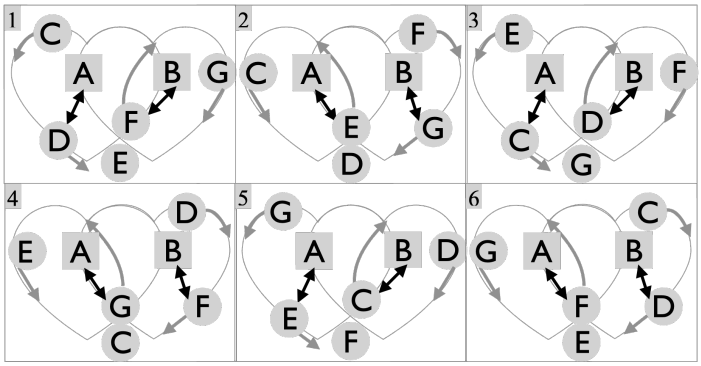
Base count		Juggler						
		A	B	C	D	E	F	G
	1	D	F		A		B	
	2	E	G			A		B
	3	C	D	A	B			
	4	G	F				B	A
	5	E	C	B		A		
	6	F	D		B		A	
	7	G	E			B		A
	8	D	C	B	A			
	9	F	G				A	B
	10	C	E	A		B		

Pattern by Peter Kasemen, conjured while thinking about ball bouncing!

Metrics: J=7, CJ=3, pH=1, PBJ=2/7, cycle=10 beats on base (4 or 6 count), R=2, OJ=20/7. Each mover covers 8 back-front segments during the cycle; if each is about ½ pu, then we get 4 pu per mover, giving pattern average motion= 20/7 pu, and spin≈ 0.

Broken Hearted Blast

When the variation is more fun than the original!



Base 4 count		Juggler						
		A	B	C	D	E	F	G
	1	D	F		A		B	
	2	E	G			A		B
	3	C	D	A	B			
	4	G	F				B	A
	5	E	C	B		A		
	6	F	D		B		A	
	7	G	E			B		A
	8	D	C	B	A			
	9	F	G				A	B
	10	C	E	A		B		

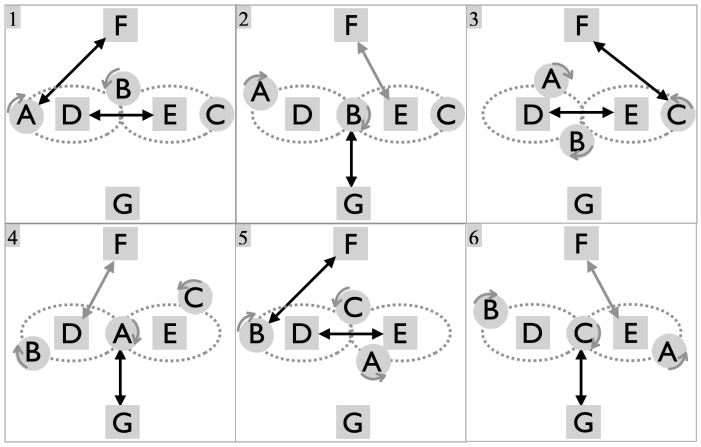
The passing table here is exactly the same as for the *Blast*, but motion is completely different in this post-Blast derivative. Instead of moving back into a stick position, the juggler walks forward between the feeders and quickly back again to complete a big loop, ending up in the same spot. It's easy to get a bit confused on who moves through when, and so we recommend doing a good run of the original *Blast* first before trying this variation. Interestingly, the necessary spin motion is very little, and movers can stay facing the same direction while following the heart-shaped path. But they cover a lot of ground!

Metrics: J=7, CJ=3, pH=1, PBJ=2/7, cycle=10 beats on base (4 count), R=2, OJ=20/7, spin≈0. Noting the dimensions of a heart shape, movers cover about 8 pu per cycle, for a pattern average 5.7 pu!

Eye of the Needle

A *Twister Weave* with Obstacles

A popular extension of the *Twister Weave* when two brave stationary jugglers want to play. Big-time movement for the weavers!



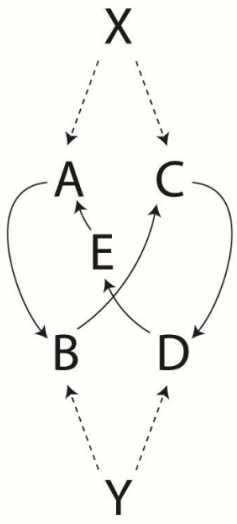
Base count	Juggler						
	A	B	C	D	E	F	G
1	**F**			**E**	**D**	**A**	
2		**G**			**F**	**E**	**B**
3			**F**	**E**	**D**	**C**	
4	**G**			**F**		**D**	**A**
5		**F**		**E**	**D**	**B**	
6			**G**		**F**	**E**	**C**
7	**F**			**E**	**D**	**A**	
8		**G**		**F**		**D**	**B**
9			**F**	**E**	**D**	**C**	
10	**G**				**F**	**E**	**A**
11		**F**		**E**	**D**	**B**	
12			**G**	**F**		**D**	**G**

Set up with **B** facing Dr. Inside (**G**), but on Dr. **F**'s side of the two needles (**D,E**). Then **G** starts self as the needles pass and moves forward passing to **G** on beat 2, and so on for other weavers moving across the **D**-**E** passing lane. Traffic can get hectic, and there's often interference between movers and the needles…all in good fun! The extra-pass version is pretty hard for non-expert needles!

Metrics: J=7, CJ=3, pH=1, cycle=12 on 2-count base, OJ=2, R=3 (standard), 4 (extra-throws), PBJ=3/14 (standard), 2/7 (extra throws). Weaver spin is about 4π rad per cycle, as in the original *Twister Weave*, and Dr. Outside (**F**) spins about 2π, as long as they are standing back a bit with longer throws to compensate for the inflated figure-8. That gives a pattern average spin $6\pi/7$ rad. (Perhaps π or a bit more for the extra-pass version.) Motion per weaver is pretty high, and requires gazelle-like finesse! We estimate it's about $\sqrt{2}$ greater than the 4 pu from the *Twister Weave* motion, for a pattern average 2.4 pu.

Twister 5

A good pattern for new movers!



This one comes by *twister-izing* the 6-person Gandini weave; that is, splitting the feed position into two (**X,Y**) and moving them to opposite sides of the weave. Thinking of the original feeder as a giant who is *juggling* the 5 movers, this version defies gravity! The standard (black font) is a 2-count right-handed base. Once mastered, there's plenty of time for even more passing by doubling up to each feeder as you move through (grey).

Base count		Juggler						
		A	**B**	**C**	**D**	**E**	**X**	**Y**
	1	**X**				Y	**A**	E
	2	X	**Y**				A	**B**
	3		Y	**X**			**C**	B
	4			X	**Y**		C	**D**
	5				Y	**X**	**E**	D
	6	**Y**				X	E	**A**
	7	Y	**X**				**B**	A
	8		X	**Y**			B	**C**
	9			Y	**X**		**D**	C
	10				X	**Y**	D	**E**

Metrics: J=7, CJ=3, pH=1, PBJ=1/7 (2/7 double time), cycle=10 beats on base (2 or 3 count), R=2, OJ=20/7. As in the six-person (Gandini 5-weave), motion per weaver is about 5 pu, for a pattern average motion ≈ 3.6 pu. Weavers spin 4π rad (2 full circles), while feeders swivel modestly, perhaps 5 sweeps each of $\pi/8$, for a pattern average spin $\approx 3\pi$ rad, or 540°.

The standard set up has a 2 count base, so feeders **X** and **Y** are on 4-count (or 2-count in the double-time version). Ignoring the double-time, speeding the pattern up is possible by having **X** and **Y** on 3-count rather than 4-count. Since the table just shows events on passing beats, we recognize that the whole thing may be described as passing events happening on beats *1-and-2-on-a-3-count*. For instance, **X** passes on every 1 and **Y** passes on every 2, and no passes on 3. This count is super nice for feeder balance and overall pace; for instance left-hand passes send the weaver off to the feeder's left and vice versa. It also leads to movers alternating 7-count-8-count as they move through their cycle. PBJ=2/21 for this *1-2 on a 3-count* version, and we would say cycle length is 30 on a 1-count; other metrics as above.

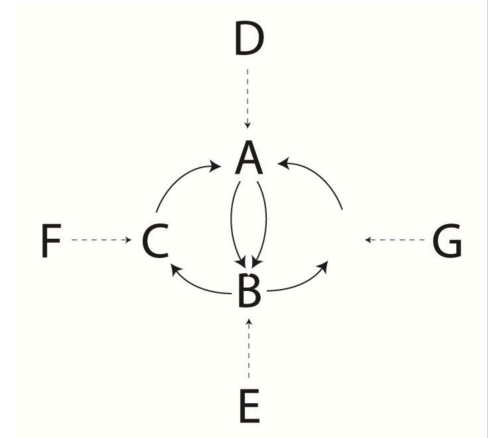
Iron Cross Weave

A weave with 4 doctors!

Weavers **A, B, C** move through figure 8 while passing to two twister-like doctors (**D,E**) and two new doctors **F** and **G**!

A good pattern for 3 busy movers and 4 happy feeders

Pattern by Luke Emery



4 count base		Juggler						
		A	**B**	**C**	**D**	**E**	**F**	**G**
	1	**D**	**E**	**F**	**A**	**B**	**C**	
	2	**E**	**G**	**D**	**C**	**A**		**B**
	3	**F**	**D**	**E**	**B**	**C**	**A**	
	4	**D**	**E**	**G**	**A**	**B**		**C**
	5	**E**	**F**	**D**	**C**	**A**	**B**	
	6	**G**	**D**	**E**	**B**	**C**		**A**

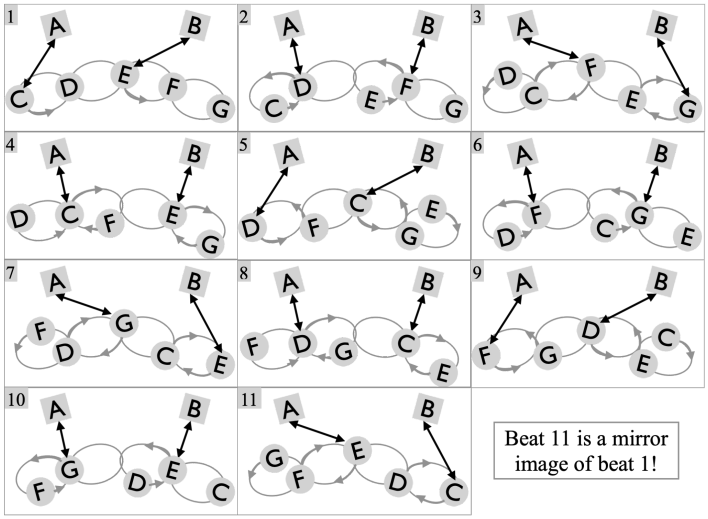
Variations: Luke recommends filling in the gaps in that 4-count base, for instance with 2 passes in a row to each doctor. Good luck!

Metrics: J=7, CJ=3, pH=1, OJ=24/7, cycle=6 on 4-count base, R=3, PBJ=3/14. Movement stats inherit from the *Twister weave* for **A**, **B, C**, giving a pattern average motion=12/7=1.7 pu. Likewise, the weavers spin 2 full circles per cycle, for a pattern average spin=$12\pi/7 \simeq 5.4$ rad.

Four-Loop Weaves

Intersecting Weaves

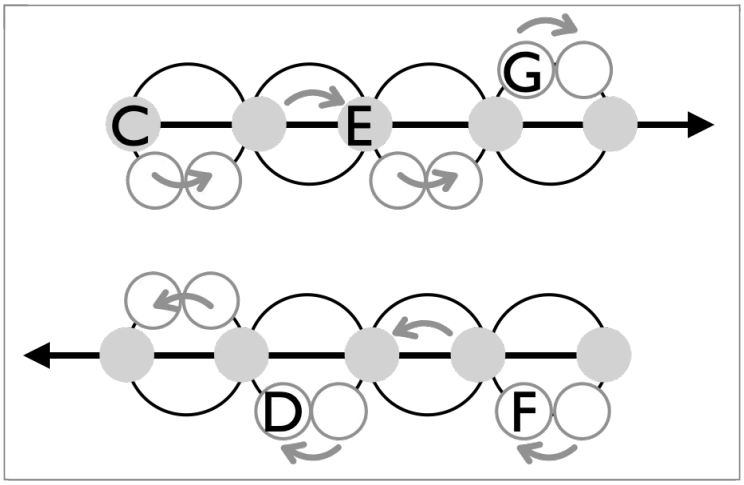
Aka *2-5 Foie Graham*



two count base	Juggler						
	A	B	C	D	E	F	G
1	C	E	A		B		
2	D	F		A		B	
3	F	G				A	B
4	C	E	A		B		
5	D	C	B	A			
6	F	G				A	B
7	G	E			B		A
8	D	C	B	A			
9	F	D		B		A	
10	G	E			B		A

Two weaves set up side by side, sharing one juggler (**E**, at first). Weavers pass a regular 4 beats on 6-count to one weaver, and then they switch to the other feeder on a quick 2-count. ½ cycle shown.

Metrics: J=7, CJ=3, pH=1, cycle=20 (on 2-count base), OJ=20/7, PBJ=2/7, R=2. Each loop is about 2 pu, giving a pattern average motion=40/7=5.7 pu. As in a *standard weave* for 4 there's not much spin, though feeders need to wobble back and forth to cover the spots. Each feeder rotates a bit less than ½ turn per back-and-forth sweep (141°, according to weave calculations), and there are 5 such sweeps per cycle. Likewise, each weaver wobbles about 55° per feeder over the cycle, giving a pattern average spin 280°.



In the steady state, weavers wobble left-to-right (top) and then right-to-left (bottom), passing at the grey spots on each line. The white circles are really phantom spots, where jugglers move through but don't pass. The five weavers are marked here where they start, consistent with the more complete figure above. Notice the grey pass spots on either end are shared by top and bottom, and notice there's a juggler at every 4th spot!

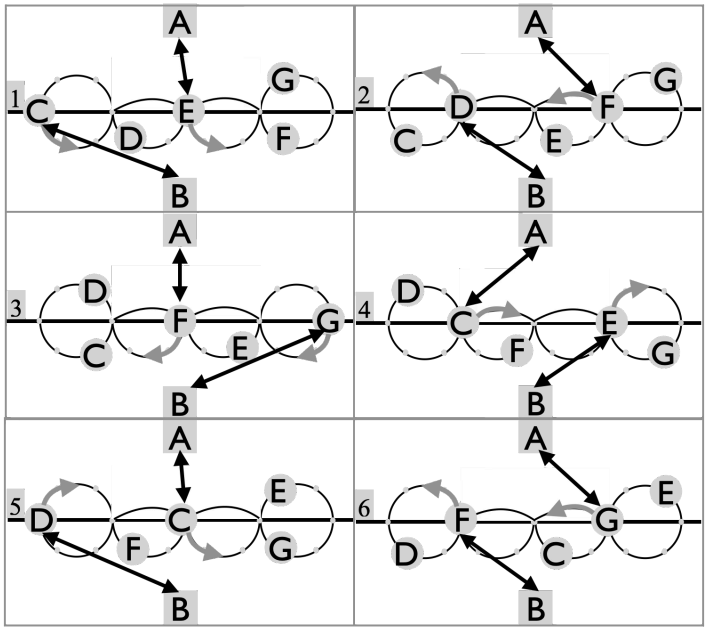
A concise description of the moving rule is to pass like in an ordinary weave (e.g., back on the end, forward through the middle, on six count), but when you are leaving a “middle” heading towards the other side of the pattern, then do two passes in a row (e.g., **F** on beats 2 and 3).

The *Intersecting Weaves* pattern is by Steve Otteson, dating perhaps to the mid-to-late 1990’s. It was “independently” rediscovered by Mellonhead as a variation of 8-person *Mel Extension of the Foie Graham,* which itself extended Graham Pasch’s 6-person *Foie Graham.* Steve’s *Intersecting Weaves* appeared in earlier versions of the Pattern Book, but we realized quite late (while preparing Version 4.0, September, 2026) that it is exactly the same as what we had later introduced and named the *2-5 Foie Graham.* What is in a name?

Mike's 1-5-1 Foie Graham

One for the ages! (Maybe the *Foie Mike*?)





2 count base		Juggler						
		A	B	C	D	E	F	G
	1	E	C	B		A		
	2	F	D		B		A	
	3	F	G				A	B
	4	C	E	A		B		
	5	C	D	A	B			
	6	G	F				B	A
	7	G	E			B		A
	8	D	C	B	A			
	9	D	F		A		B	
	10	E	G			A		B

Notice **A** just feeds edges of the inner two loops, while **B** feeds the edges of the outer two loops.

Twisterize the previous pattern and reimagine roles of the feeders, which is necessary to deal with switching sides and 2-count, and we get this beautiful new weave. For pretty advanced weavers.

Weavers proceed through the 4 loops (upper right). The passing rule is straightforward; when heading into an inner loop, pass twice (say on 2-count) to **A.** Treating that sequence like the old "middle" of a standard weave, then you leave the middle, reverting to 6-count with **B,** which you do for two passes before re-entering the middle.

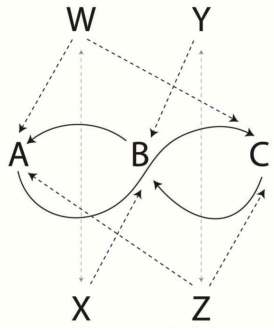
Metrics: J=7, CJ=3, pH=1, cycle=20, on 2-count base, OJ=20/7, PBJ=2/7, R=3. Motion = 5.7 pu inherits from the *2-5 Foie Graham*, but there's much more spin here to handle the direction changes. Each weaver spins about 4 full turns and each feeder makes 5 sweeps per cycle, for pattern average spin of 19.1 rad! The original 1-5-1 had a different and less appealing pass sequence, and we probably only juggled it once when Graham was living in Madison. The version here was born September 20, 2026.

Extra Club Moving Seven Person Patterns

Twister Plus Plus

A big deal!

This amazing pattern is another crown jewel of the MAJ pattern repertoire. It's a combination of *Twister Plus* and *Double Time Twister*. The feeders operate as in *Twister Plus*, alternating triples across the pattern with singles to the weavers. The weavers do a *Double Time Twister Weave,* passing twice on each side to the two feeders, and then changing direction on 1 right-hand self. Not for the faint of heart!



Drs. Inside: **X**, **Y**
Drs. Outside: **W**, **Z**

	Beat (2-count base)											
Juggler	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**
A	**W**	**Y**	⤴	**X**	**Z**	⤴	**W**	**Y**	⤴	**X**	**Z**	⤴
B	⤴	**X**	**Z**	⤴	**W**	**Y**	⤴	**X**	**Z**	⤴	**W**	**Y**
C	**Z**	⤴	**W**	**Y**	⤴	**X**	**Z**	⤴	**W**	**Y**	⤴	**X**
W	**A**	t-**X**	**C**	t-**X**	**B**	t-**X**	**A**	t-**X**	**C**	t-**X**	**B**	t-**X**
X	t-**W**	**B**	t-**W**	**A**	t-**W**	**C**	t-**W**	**B**	t-**W**	**A**	t-**W**	**C**
Y	t-**Z**	**A**	t-**Z**	**C**	t-**Z**	**B**	t-**Z**	**A**	t-**Z**	**C**	t-**Z**	**B**
Z	**C**	t-**Y**	**B**	t-**Y**	**A**	t-**Y**	**C**	t-**Y**	**B**	t-**Y**	**A**	t-**Y**

The inside doctors, **X** and **Y,** start with 4 clubs each, leading the pattern by launching triples over the top.

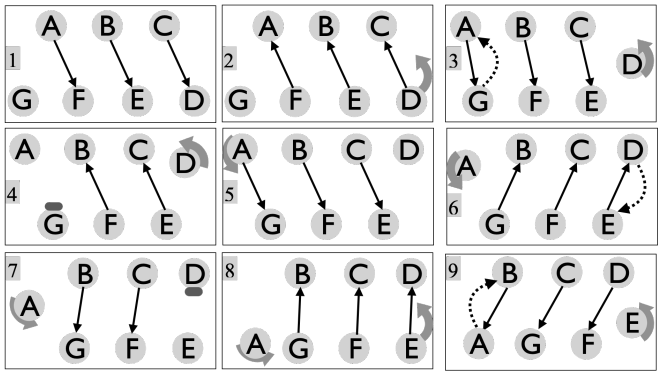
Pattern by MAJ.

Metrics: J=7, OJ=3.4, CJ= 3.3 (i.e., 2 extra), R=2, PBJ=¾ (that's high!), cycle length=12 on 2-count base. pH=3/2 (averaging triples and singles). Motion 12/7 pu, noting that weavers cover 4 pu per cycle. Spin per weaver is about like the *twister weave*, at 2 full turns per cycle; the outside doctors cover a fair swath; they need to be close enough to weave line so their triples are manageable, which makes for somewhat bigger angles. We estimate ¾ of a full turn per doctor per cycle, for a pattern average spin $2.6\pi \simeq 8$ rad.

Big Mangled Squirrel

Everybody in the action!
Triple 2-1-2-1-2

An exciting extension of the 5 person *Mangled Squirrel* (aka *Mango Swirl*), formed by extending the 2 v 3 set up to a 3 v 4. All spots have the same pass/move sequence and are all equally difficult! Jugglers **A**, **B**, **C**, and **G** are in phase to start, with two clubs each in their right hands. Jugglers **A, B**, and **C** start with 4 clubs each. Jugglers **D**, **E**, and **F** have 3 clubs each, and each start with 2 in the left.



All passes are straight right doubles, except for the one triple pass after you turn, followed by a pause to bring the mover into phase with the line they join. The steady state rule per juggler is: move, triple pass, pause, 2 doubles to one, 1 double to the next, 2 doubles to the next, 1 to the next, and 2 to the last; that is, *Triple-2-1-2-1-2*. In the table, "↺" is the change side symbol, set in a shaded box sequence, and "–" is the pause symbol. All passes are right doubles, except for the right triples indicated with `t`. As noted above, jugglers' hands are out of phase. To lock in on who receives from you twice vs once, form up in a circle. The person on your left will get the triple, the next person gets two doubles, the next one double, and so on. Generally, if you pass someone 2 doubles, they pass you just one, and *vice versa*. (Note **C**'s pass on beat 1 is a starting configuration only; in the steady state they would have passed a triple 1 beat earlier, as in line 21)

Beat 1-cnt	Juggler						
	A	**B**	**C**	**D**	**E**	**F**	**G**
1	**F**	**E**	**D**				
2				**C**	**B**	**A**	
3	**G**	**F**	**E**				t-**A**
4				↺	**C**	**B**	**–**
5	**G**	**F**	**E**				
6				t-**E**	**D**	**C**	**B**
7	↺	**G**	**F**	**–**			
8					**D**	**C**	**B**
9	t-**B**	**A**	**G**	**F**			
10	**–**				↺	**D**	**C**
11		**A**	**G**	**F**			
12	**C**				t-**F**	**E**	**D**
13		↺	**A**	**G**	**–**		
14	**C**					**E**	**D**
15		t-**C**	**B**	**A**	**G**		
16	**D**	**–**				↺	**E**
17			**B**	**A**	**G**		
18	**E**	**D**				t-**G**	**F**
19			↺	**B**	**A**	**–**	
20	**E**	**D**					**F**
21			t-**D**	**C**	**B**	**A**	↺

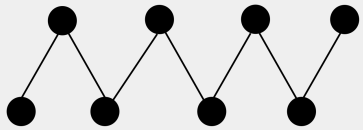
Metrics: J=7, CJ=3.4, cycle=21, base=1-count, R=1, PBJ=3/7, OJ=6, pH=2 1/9, R=1. Motion and spin inherit from the 5-person pattern, at about: spin = 2π rad, motion $\simeq 1$ pu.

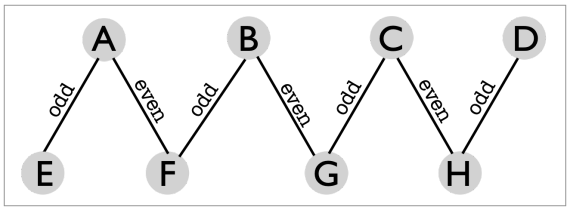
Eight Person Patterns

Stationary 8

Simple 4 v 4

For beginners.





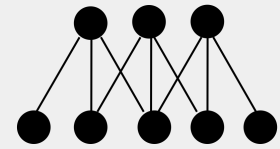
Baby positions **D** and **E**.

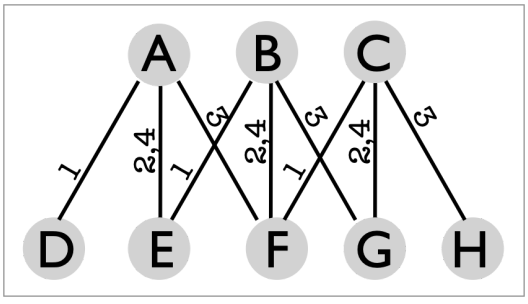
Base beat		Juggler							
		A	B	C	D	E	F	G	H
	1	E	F	G	H	A	B	C	D
	2	F	G	H			A	B	C

Metrics: J=8, OJ=7/4, CJ=3, R=2, PBJ=(⅞)x(1/base), cycle=2 on base count, pH=1,motion=0, spin=0.

Simple 3 v 5

For beginning feeders





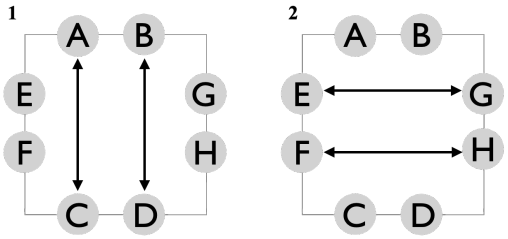
Base beat		Juggler							
		A	B	C	D	E	F	G	H
	1	D	E	F	A	B	C		
	2	E	F	G		A	B	C	
	3	F	G	H			A	B	C
	4	E	F	F		A	B	C	

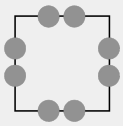
…comment on base beat…usually 2-count..

Metrics: J=8, OJ=2 ¼ , CJ=3, R=3, PBJ=(¾)x(1/base), cycle=4 on base beat, pH=1, motion=0, spin=0.

Double Vision Box

Collision risk scares some beginners!





4 count base		Juggler							
		A	B	C	D	E	F	G	H
	1	C	D	A	B				
	2					G	H	E	F

Easy, in principle, but requires great timing with long crisp throws, so it's not for everyone!

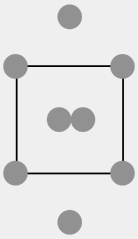
Metrics: J=8, OJ=1, CJ=3, R=1, PBJ=1/base, cycle=2 on base count, pH=1, motion=0, spin=0.

Big Hinge (minus 4 centers)

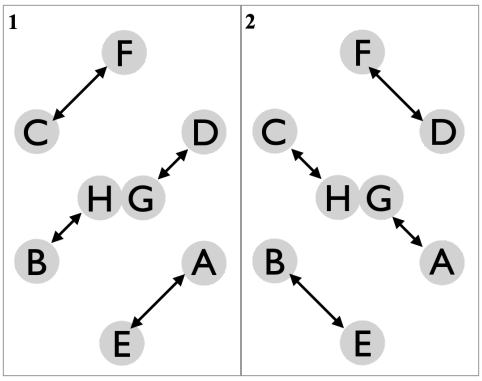
The word to remember…**swivel**!

Beat (3 or 4-count base)

A good short-cycle stationary pattern that's simple to set up. It requires a lot of rotation in place to face up to your partner.



Juggler	1	2
A	E	G
B	H	E
C	F	H
D	G	F
E	A	B
F	C	D
G	D	A
H	B	C



Pattern arises from the 12-person *Big Hinge* by removing the 4 baby positions! In that parent pattern, **E**, **F, G**, and **H** are feeders, and **A**, **B**, **C**, **D** are hinges.

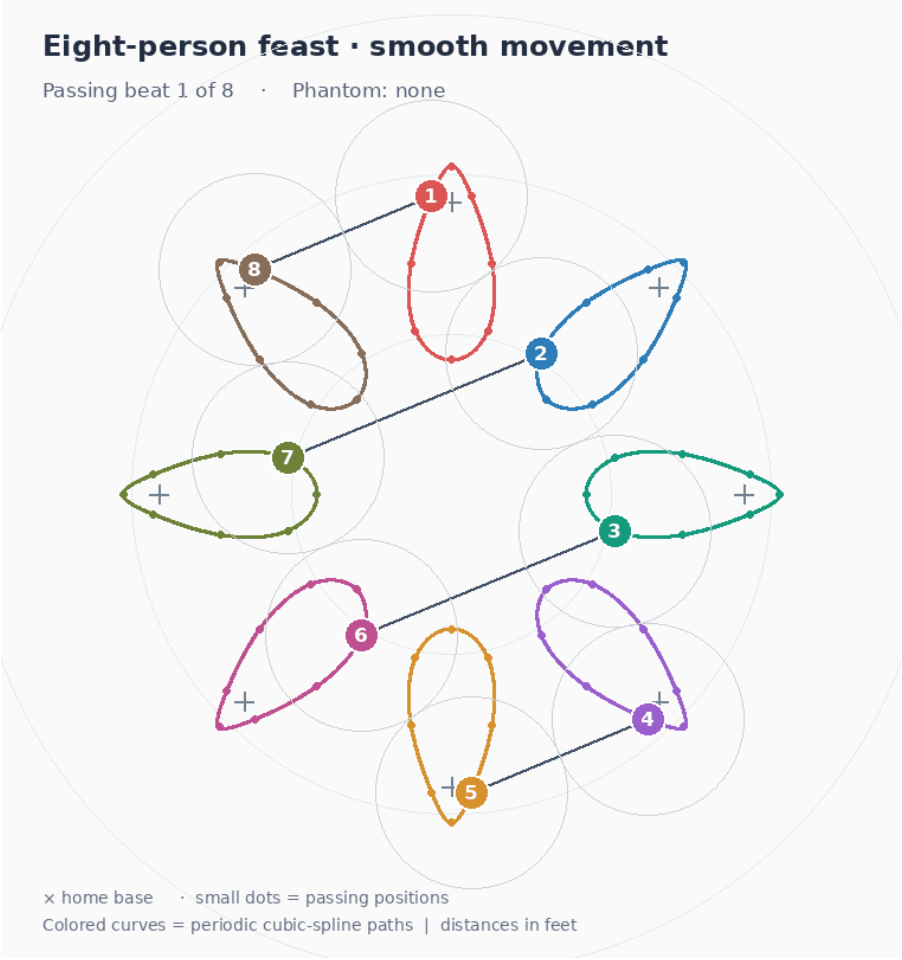
Metrics: J=8, OJ=2, CJ=3, R=1, PBJ=1/base, cycle=2 on base count (3 or 4 works fine). pH=1, motion=0, spin=π rad.

Extra-Club Stationary 8

?

Moving 8

Benzene Ring
1-3-3-1 Twister
Mating 3-leaf clovers
Double vision 3-leaf clover
Polymer weaves
Double vision shooting star
Double vision oogle boogie might not be possible without switching teams!
Khaos Butterfly
Egg-beater twister; standard + everyone moves
3-5 Foie Graham….isnt' this Mel extension of Foie Graham
Mel's Dream; original; naturalized



Extra Club Moving 8

Extra Club Inverted Benzene Ring

Nine Person Patterns

Ten Person Patterns

Eleven Person Patterns

Twelve Person Patterns

Big Patterns

Patterns for an indefinite number of jugglers!

…an idea whose time has come…like

Pattern Statistics

Pattern	N Jugglers	N clubs / J	PBJ	N moves / cycle / Jug	N rotations / Cycle / jug	SD ?
